Adversarial Attack and Defense for Machine Learning in Statistical Physics

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Abstract

Recently, various machine learning (ML) systems have been proposed for physical science. Their powerful mathematical expressiveness offers an inherent advantage in maintaining the desirable statistical properties of physics. However, they are placed in ideal and virtual environments that seldom consider noise, perturbations, or various real-world experimental limitations. This exposes significant vulnerabilities in several aspects: 1) The input of an ML system can be subject to human or natural, deliberate or unconscious perturbations, which may sabotage the running baseline of the ML system. 2) A small input perturbation may be amplified by the maximum physical sensitivity, which may lead to an extreme output or a divergent loss function value. 3) A perturbed input may violate some strict statistical assumptions that the ML system relies on, which may lead to incorrect scientific results and findings. In this work, we propose a complete attack-defense methodology for ML systems in statistical physics. In the attack side, we launch a black-box attack (such as a small Gaussian noise) at the input, which stays within physically plausible limits and preserves physical properties. In the defense side, we develop a functional local variation regularized learning scheme to capture and suppress the adversarial perturbation. Theoretical analysis and extensive experiments show the effectiveness of the proposed method. This finding may improve the robustness of macroparameter estimation for such ML systems via adversarial optimization.

1 Introduction

With the rapid development of machine learning (ML) systems in recent years, various new models and usages have been proposed for different fields of physical science. For example, the additivity of Brownian motion (a statistical property) can be exploited to sum up the reactive forces associated with the edges in an NN [5]. A distance-aware uncertainty estimator is developed to model the dynamics of Markov decision processes [29]. The Markov Chain Monte Carlo (MCMC) sampling on spin configurations can be used to calculate the expectation values of Hamiltonians [22].

Although such ML systems have pushed the frontiers of physical science, they seldom consider noise, perturbations, or various real-world experimental limitations. This has left behind significant problems with systemic vulnerability and scientific security. ❶ The inputs of such ML systems may be directly observed from physical experiments or generated through specialized large-scale computer simulations, which may contain human or natural, deliberate or unconscious perturbations

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[4, 55, 3, 13, 28, 7]. ❷ The physical processes embedded in ML systems can be highly sensitive to input perturbations, which may yield extreme or incorrect outputs [19, 18, 45, 51, 59, 8, 63]. ❸ Adversarially-perturbed inputs may violate strict statistical assumptions and lead to incorrect scientific results. The smoothness or local robustness assumption in computational materials science can be violated by non-physical discontinuities in the decision boundary [64, 61, 52, 56, 44, 34].

We present several approaches that an attacker may take to exploit the above vulnerabilities. ❶ A black-box attack at the input with even mean-zero Gaussian noise can sabotage the running baseline of an ML system. **In this case, the good statistical properties of an ML system can also provide good concealment for adversarial attacks.** As the number of samples approaches infinity, while an estimator consistently converges to some value, the cumulative adversarial noise may simultaneously converge to zero and make it difficult to detect (see Eqs. 7, 8, and 27). ❷ The attacker can target ML systems that are highly physically sensitive due to their theoretical nature. An ML system with a high condition number, such as a Physics-Informed NN (PINN), can be hard to train and obtain unsatisfactory accuracy [16]. Once attacked by a perturbed input, such ML systems may produce an extreme output or a divergent loss function value. ❸ The attacker can explore strict statistical assumptions that can be easily violated by a perturbed input to yield incorrect scientific results. The standard empirical risk minimization training scheme assumes that the training and test data are drawn independently and identically distributed (i.i.d.) from the same underlying distribution, which can be easily violated by distributional shifts [31, 57] or carefully crafted adversarial examples [20, 40, 9] and leads to misclassifications or incorrect results.

Given the above vulnerabilities and threats, it is urgent to develop a suitable defense methodology for ML systems in statistical physics. We notice that total variation (TV) is a popular approach that quantifies the perturbation propagation from the input to the objective function [49, 10, 12, 11, 31, 57, 32]. However, it is challenging to generalize the classical concept of TV from discrete pixel grids to a more abstract functional space. Besides, since such ML systems employ only one fixed set of physical features, we need to construct the local variation (LV) [33] in our task, which corresponds to a cutting plane of TV. To overcome these challenges, we develop a functional LV (FLV) that can quantify the LV w.r.t. the physical features. Its key technique is to employ both functional analysis and variational analysis [47] in the domain of statistical physics. When using this FLV as regularization in the learning process, the perturbation of loss function induced by the distortion of physical features can be suppressed. Our main contributions can be summarized as follows:

1. We propose a new threat model that targets ML systems in statistical physics. It stays within physically plausible limits and preserves physical properties, which makes it inconspicuous.

2. We propose an FLV regularized learning scheme to capture and suppress the adversarial perturbation, which forms an Adversarial Defense scheme for ML systems in Statistical Physics (**ADSP**).

3. Since the proposed FLV is non-differentiable, we further develop a subdifferential calculation scheme for ADSP, which is compatible with backpropagation and mainstream deep learning architectures like Pytorch[2].

The field of statistical physics is vast, making an exhaustive analysis of its many ML systems intractable. However, these systems often share the above-mentioned common vulnerabilities. Hence we use a recent ML system on a standard 2D Ising model [22] as a representative to present our methodology. We also extend it to a 3-state Potts model [46] by using the corresponding physical background knowledge, which shows the generalizability of the threat model and FLV defense.

2 Background

2.1 Preliminaries for the Target ML System on 2D Ising Model

Before introducing the proposed methodology, we provide some necessary preliminaries for the target ML system [22] on 2D Ising Model and the corresponding background physical knowledge. To improve readability and ease comprehension, we have simplified some overly abstruse details without loss of generality. Consider a statistical physics model with a Hamiltonian functional $H[\beta, x]$, where β and x represent macroparameters and microstates, respectively. To be more specific, let $x \in \{\pm 1\}^{d^2}$ (d^2-dimensional) be the spins $x_i = \pm 1$ living on the sites of a square lattice at temperature $1/\beta$ in a 2D Ising model [25]. Then the corresponding Hamiltonian is represented by

$$H[\beta, x] = -\beta \sum_{\langle ij \rangle} x_i x_j, \tag{1}$$

where the sum $\sum_{\langle ij \rangle}$ is conducted over all the neighboring links $x_i x_j$.

[2] `https://pytorch.org/`

Note that in statistical physics, it is generally impossible to simultaneously observe every single sample x. Instead, we can only observe bulk properties or macroscopic observables $\{O_\alpha\}_\alpha$ (with α being the index that labels different observables) derived from the full samples $\{x\}$. For each observable $O_\alpha(x)$, we can actually observe its expectation value under the Hamiltonian H:

$$\langle O_\alpha(x)\rangle_H = \frac{1}{Z}\sum_x O_\alpha(x)\exp(-H[\beta, x]), \tag{2}$$

where $Z = \sum_x \exp(-H[\beta, x])$, and $\exp(-H[\beta, x])/Z$ serves as a distribution function for x under this Hamiltonian.

The next step is to extract some key physical features $\{\phi^a\}_a$ (with a being the label index) from the raw observables $\{O_\alpha\}_\alpha$, such as traditional energy/coupling terms, local angular correlations, intermediate-range correlations, and local gauge invariant quantity (see Table A1). These features contain important physical structural information, which plays a crucial role in the ML system. Then the expectations of observables in (2) are mapped to the expectations of features:

$$\{\langle O_\alpha(x)\rangle_H\}_\alpha \mapsto \{\langle \phi^a(x)\rangle_H\}_a, \tag{3}$$

which are the main items processed in the ML system.

2.2 Task for the ML System

Given a set of features $\{\langle \phi^a(x)\rangle_H\}_a$ observed from a fixed environment with unknown macroparameters β, the task for the ML System is to retrieve any one of the following three solutions: the original solution $\tilde{\beta}^* = \beta$, the antiferromagnetic analogue $\tilde{\beta}^* = -\beta$, or the dual solution $\tilde{\beta}^*$ such that $\sinh(2\tilde{\beta}^*)\sinh(2\beta) = 1$ [17]. To perform this task, one can try any macroparameters $\tilde{\beta}$ to generate a new set of samples $\tilde{x}$ and a new Hamiltonian $\tilde{H}$ in the same environment, and use a sufficiently flexible NN G to recover all the relevant lattice models to the original one. In the rest of this paper, we denote a regenerated term by adding a tilde ($\tilde{x}$) on top. Then a new set of expectations of the physical features $\{\langle \phi^a(G(\tilde{x}))\rangle_{\tilde{H}}\}_a$ can be obtained. Based on the duality theory [30], if $\tilde{\beta}^*$ is one of the above three solutions, then

$$\langle \phi^a(x)\rangle_H = \langle \phi^a(G(\tilde{x}^*))\rangle_{\tilde{H}^*}, \quad \forall a. \tag{4}$$

Based on this theory, the following learning model can be established:

$$\arg\min_{\tilde{\beta},\theta}\left\{\mathcal{L}(\tilde{\beta},\theta) := \sum_a (\langle \phi^a(x)\rangle_H - \langle \phi^a(G_\theta(\tilde{x}))\rangle_{\tilde{H}})^2\right\}, \tag{5}$$

where θ contains the learnable parameters for the NN G. An alternating direction scheme (e.g., [39, 38]) can be used to learn $\tilde{\beta}$ and θ. The gradient $\partial_\theta \mathcal{L}$ can be calculated by a conventional automatic differential technique like `autograd`, while $\partial_{\tilde{\beta}}\mathcal{L}$ can be calculated with the observables $\{\langle O_\alpha(\tilde{x})\rangle_{\tilde{H}}\}_\alpha$ and the expectations $\{\langle \phi^a(G(\tilde{x}))\rangle_{\tilde{H}}\}_a$. Further details that are peripheral to this study can be found in [22].

3 Methodology

Now we turn to present our methodology from the attack side to the defense side in the above target ML system on 2D Ising model. In brief, the attack is launched at the input x (experiments on attacking the features $\{\phi^a\}_a$ are also conducted in Section 4.3), while the defense is deployed in the learning model (5). In this way, the attack and defense form a complete closed loop, from which we can gain some key insights.

3.1 Attack Side

The key attack technique is to exploit the intrinsic vulnerability in statistical physics that one can only observe macroscopic observables $\{O_\alpha\}_\alpha$ derived from a sufficiently large number of samples $\{x\}$, instead of simultaneously observing every single sample x. Whether these samples are obtains from physical experiments or generated by simulations, they can be subject to human or natural, deliberate or unconscious perturbations. For example, Johnson-Nyquist noise is a typical noise in thermodynamics, let alone the Gaussian noise that is ubiquitous in various detectors, measurements, and nature. However, the target ML system lacks robustness against such noises, making it susceptible to adversarial attacks.

Stealth of the Attack. We show that even small Gaussian noise injected into the samples can significantly degrade the target system performance. Moreover, this Gaussian noise can be effectively concealed by the aggregate operation of multiple samples. Specifically, let mean-zero Gaussian

noise $\delta_{(n),ij} \sim \mathcal{N}(0, \epsilon^2)$ be identically and independently sampled and injected into each clean link $x_{(n),i}x_{(n),j} \in \{\pm 1\}$ to yield a perturbed link

$$x'_{(n),i}x'_{(n),j} = x_{(n),i}x_{(n),j} + \delta_{(n),ij}, \quad n = 1, \cdots, N, \tag{6}$$

where $\epsilon > 0$ is the attack budget. In the rest of this paper, we denote an adversarially-attacked sample by using a superscript prime ($'$). *One may argue that $x'_{(n),i}x'_{(n),j}$ may violate the physical feasibility constraint $\in \{\pm 1\}$ and thus it can be easily detected and removed. However, one single $x'_{(n),i}x'_{(n),j}$ cannot be observed individually in statistical physics, especially for strongly interacting lattice systems represented by the Ising model, thus the defender cannot examine whether $x'_{(n),i}x'_{(n),j} \in \{\pm 1\}$. Instead, the average link observable over the full samples is*

$$\begin{aligned} O'_{link} &= \frac{1}{N|\{\langle ij\rangle\}|}\sum_{n,\langle ij\rangle} x'_{(n),i}x'_{(n),j} = \frac{1}{N|\{\langle ij\rangle\}|}\sum_{n,\langle ij\rangle}(x_{(n),i}x_{(n),j} + \delta_{(n),ij}) \\ &= \frac{\sum_{n,\langle ij\rangle} x_{(n),i}x_{(n),j}}{N|\{\langle ij\rangle\}|} + \frac{\sum_{n,\langle ij\rangle}\delta_{(n),ij}}{N|\{\langle ij\rangle\}|} \qquad (7) \\ &\approx \frac{\sum_{n,\langle ij\rangle} x_{(n),i}x_{(n),j}}{N|\{\langle ij\rangle\}|} + 0 = O_{link}. \qquad (8) \end{aligned}$$

Since $\{\delta_{(n),ij}\}_{\langle ij\rangle}$ are mutually independent mean-zero noises, as $N|\{\langle ij\rangle\}| \to \infty$, the second term in (7) equals

$$\frac{\sum_{n,\langle ij\rangle}\delta_{(n),ij}}{N|\{\langle ij\rangle\}|} \to \mathbb{E}[\delta_{(n),ij}] = 0. \tag{9}$$

Therefore, (8) *indicates that $O'_{link} \approx O_{link}$, which seems that the Gaussian noise "has vanished". More importantly, averaging the clean links $\{x_{(n),i}x_{(n),j}\}_{\langle ij\rangle} \subseteq \{\pm 1\}^{|\{\langle ij\rangle\}|}$ into the clean average observable $O_{link} \in \mathbb{R}^{|\{\langle ij\rangle\}|}$ has relaxed the physical feasibility constraint $\{\pm 1\}^{|\{\langle ij\rangle\}|}$ to the general real space $\mathbb{R}^{|\{\langle ij\rangle\}|}$. Thus the perturbed average observable O'_{link} remains within the feasibility set of O_{link}, and is very close to it. These properties subtly conceal the attack.*

Interference with the Training Process. Though subtly concealed, the Gaussian noise has already sabotaged the running baseline of the target system. For example, the regenerated perturbed samples $\{\tilde{x}'\}$ are fed into the NN G_θ in (5), which may cause distortions since NNs are widely recognized to be sensitive to input perturbations. Besides, the target G_θ is designed to have the attention mechanism, which can be perturbed by the independent noises $\{\tilde{\delta}_{(n),ij}\}_{\langle ij\rangle}$ because these noises increase the difficulty for the system to distinguish the statistical differences between different attention embeddings (i.e., different link features). Specifically, for each perturbed sample $\tilde{x}' \in \mathbb{R}^{d^2}$, the system first extracts all the neighboring links as $\tilde{x}'_{\langle ij\rangle} \in \mathbb{R}^{2d^2}$, then further extracts 7 local first-order link features for each link (see Figure A1), forming a whole feature set $\tilde{y}'_{\langle ij\rangle} \in \mathbb{R}^{2d^2\times 7}$. The target G_θ involves three learnable parameters $\theta_1 \in \mathbb{R}^7$, θ_2, $\theta_3 \in \mathbb{R}$:

$$G_\theta(\tilde{x}') = \theta_2 \cdot (\tilde{y}'_{\langle ij\rangle}\mathcal{G}(\theta_1)) + \theta_3, \tag{10}$$

where $\mathcal{G} : \mathbb{R}^7 \mapsto \mathbb{R}^7$ denotes the Gumbel-Softmax function [24] that calculates the attention weights. *Note that G_θ is designed only to establish relevant lattice models for the regenerated samples to the original lattice model, which mimics a physical transformation that can neither examine whether $\tilde{x}' \in \{\pm 1\}^{d^2}$ nor restore $\tilde{x}'$ to a clean $\tilde{x}$. Hence $\tilde{x}'$ evades this check (see Figure A1).* Since $\tilde{y}'_{\langle ij\rangle}$ are first-order link features, we have $\tilde{y}'_{\langle ij\rangle} = \tilde{y}_{\langle ij\rangle} + \tilde{\delta}_{\langle ij\rangle}$. Then (10) becomes

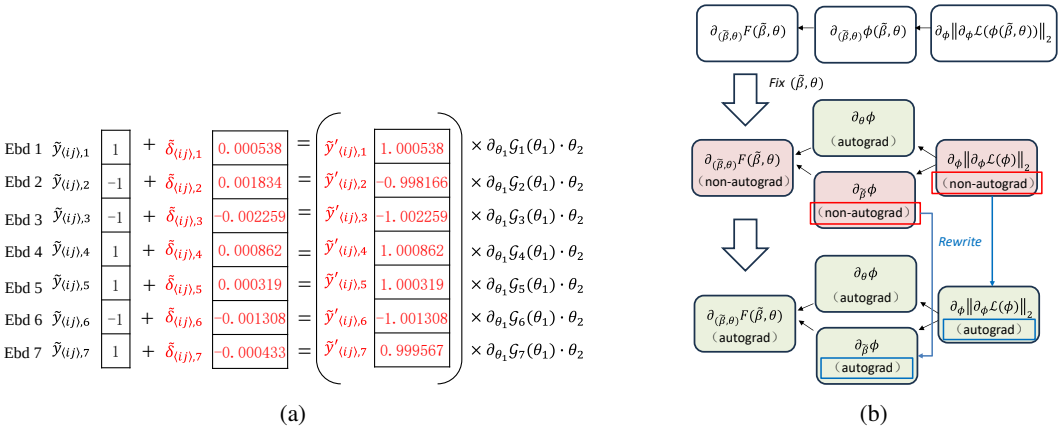
$$G_\theta(\tilde{x}') = \theta_2 \cdot (\tilde{y}_{\langle ij\rangle}\mathcal{G}(\theta_1)) + \theta_3 + \underline{\theta_2 \cdot (\tilde{\delta}_{\langle ij\rangle}\mathcal{G}(\theta_1))}, \tag{11}$$

which indicates that the forward operation involves a perturbation $\theta_2 \cdot (\tilde{\delta}_{\langle ij\rangle}\mathcal{G}(\theta_1))$. This induced perturbation can amplify the original perturbation $\tilde{\delta}_{\langle ij\rangle}$ by a factor of at most $|\theta_2|\|\mathcal{G}(\theta_1)\|_2$.

As for the backward operation (backpropagation), the partial gradient of G_θ w.r.t. θ_1 is

$$\partial_{\theta_1} G_\theta(\tilde{x}') = \tilde{y}_{\langle ij\rangle}\partial_{\theta_1}\mathcal{G}(\theta_1) \cdot \theta_2 + \underline{\tilde{\delta}_{\langle ij\rangle}\partial_{\theta_1}\mathcal{G}(\theta_1) \cdot \theta_2}. \tag{12}$$

Originally, the 7 attention heads should have concentrated only on the 7 ground-truth local features $\tilde{y}_{\langle ij\rangle}$. However, since $\tilde{\delta}_{\langle ij\rangle}$ are i.i.d. perturbations, they impose random interference on the multipli-



(a) (b)

Figure 1: **(a)** Diagram of how the injected perturbations $\tilde{\delta}_{\langle ij \rangle}$ interfere with the attention mechanism (7 attention embeddings). The i.i.d. perturbations $\{\tilde{\delta}_{\langle ij \rangle, k}\}_{k=1}^{7}$ weaken the distinctions between different features $\{\tilde{y}_{\langle ij \rangle, k}\}_{k=1}^{7}$, which interferes with the multiplication of learning steps $\{\partial_{\theta_1} \mathcal{G}_k(\theta_1)\}_{k=1}^{7}$. This may prevent the target system from attending to the critical feature heads. **(b)** The entire calculation of $\partial_{(\tilde{\beta}, \theta)} F(\tilde{\beta}, \theta)$. The first step is to fix $(\tilde{\beta}, \theta)$ and differentiate $F(\phi) = \|\partial_\phi \mathcal{L}(\phi)\|_2$ w.r.t. ϕ. The second step is to rewrite the two non-differentiable modules $\partial_\phi \|\partial_\phi \mathcal{L}(\phi)\|_2$ and $\partial_{\tilde{\beta}} \phi$. The third step is to use the `autograd` scheme to calculate $\partial_{(\tilde{\beta}, \theta)} F(\tilde{\beta}, \theta)$ and learn the parameters $(\tilde{\beta}, \theta)$.

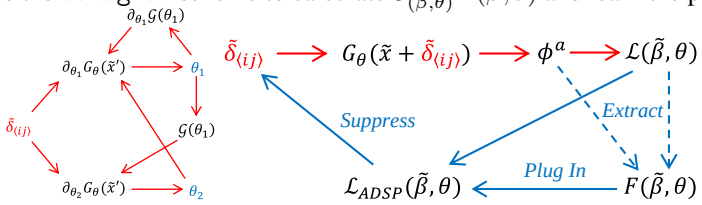


Figure 2: **Left**: the attack chain of adversarial perturbations $\tilde{\delta}_{\langle ij \rangle}$ at the learnable parameters θ_1 and θ_2 in the target system. **Right**: the defense strategy ADSP first extracts the FLV $F(\tilde{\beta}, \theta)$ from the key physical features ϕ^a and the base loss $\mathcal{L}$, then plugs it in the total loss $\mathcal{L}_{ADSP}$. Minimizing $\mathcal{L}_{ADSP}$ can suppress the adversarial perturbations $\tilde{\delta}_{\langle ij \rangle}$ by reducing the random fluctuation of $\mathcal{L}$ w.r.t. ϕ^a.

cations of learning steps $\partial_{\theta_1} \mathcal{G}(\theta_1)$ w.r.t. the 7 different attention embeddings. Consequently, $\tilde{\delta}_{\langle ij \rangle}$ weaken the distinctions between different features, thus preventing the target system from attending to the critical feature heads (see Figure 1a).

Since $\mathcal{G}$ is a Gumbel-Softmax function, $\partial_{\theta_1} \mathcal{G}(\theta_1)$ can be sensitive to θ_1, which can amplify the effect of $\tilde{\delta}_{\langle ij \rangle}$ on θ_1 by a factor of at most $\|\partial_{\theta_1} \mathcal{G}(\theta_1)\|_2$. Moreover, the perturbations are transmitted to the succeeding gradients $\partial \phi^a / \partial G_\theta$ and $\partial \mathcal{L} / \partial \phi^a$, which further amplifies the distortion. Worse still, different parameters (i.e., θ_1 and θ_2) can also interfere with each other. The partial gradient of G_θ w.r.t. θ_2 is

$$\partial_{\theta_2} G_\theta(\tilde{x}') = \tilde{y}_{\langle ij \rangle} \mathcal{G}(\theta_1) + \underline{\tilde{\delta}_{\langle ij \rangle} \mathcal{G}(\theta_1)}, \tag{13}$$

which can be jointly perturbed by both $\tilde{\delta}_{\langle ij \rangle}$ and $\mathcal{G}(\theta_1)$. In this case, $\tilde{\delta}_{\langle ij \rangle}$ perturbs θ_1, $\mathcal{G}(\theta_1)$, $\partial_{\theta_2} G_\theta(\tilde{x}')$, and finally θ_2. Conversely, θ_2 perturbs $\partial_{\theta_1} G_\theta(\tilde{x}')$ by (12), and the subsequent θ_1. This mechanism forms a persistent attack chain (see Figure 2), which may sabotage the running baseline of the target system. The attack chain of $\tilde{\delta}_{\langle ij \rangle}$ at the learnable parameter $\tilde{\beta}$ is similar to the above analysis, which is omitted here.

3.2 Defense Side

In Section 3.1, we illustrate how even small Gaussian noise can sabotage the target ML system. Since Gaussian noise is unavoidable in various detectors, measurements, natural environments, and simulations, the adversarial attack poses a significant challenge for the defense side.

The key defense technique is to exploit the randomness of Gaussian noise, which is exactly its advantage in the attack side. The analysis in Section 3.1 indicates that the i.i.d. perturbations $\tilde{\delta}_{\langle ij \rangle}$ are

transmitted to the subsequent terms G_θ, ϕ^a, and finally the base loss $\mathcal{L}$. Since these perturbations are random, the distortions of G_θ, ϕ^a, and $\mathcal{L}$ caused by them are also random. Such random distortions could be discriminated from the intrinsic physical features, because the latter are generally ordered and inherent. Based on this motivation, we construct the following FLV

$$F(\tilde{\beta}, \theta) := \|\partial_{\phi^a} \mathcal{L}(\tilde{\beta}, \theta)\|_2 \tag{14}$$

to extract the distortion of base loss $\mathcal{L}$ caused by the distortion of key physical features ϕ^a. Since the target system employs only one fixed set of physical features ϕ^a, we construct LV in our task, which corresponds to a cutting plane of TV at ϕ^a. The reason for differentiating w.r.t. ϕ^a is that ϕ^a contain important physical structural information, such as traditional energy/coupling terms, local angular correlations, intermediate-range correlations, and local gauge invariant quantity. Hence the distortion of ϕ^a has a significant negative impact on the fidelity of $\mathcal{L}$. Besides, since ϕ^a are observable features, it eases further investigations in the reliability and trustworthiness of such ML systems and physical experiments.

The FLV has a good property of capturing random perturbation propagation from ϕ^a to $\mathcal{L}$. In the perspective of defense, it can be suppressed to reduce the random fluctuation of $\mathcal{L}$ w.r.t. ϕ^a. Thus we propose the following FLV regularized learning scheme as an Adversarial Defense strategy for Statistical Physics (ADSP):

$$\arg\min_{\tilde{\beta}, \theta} \left\{ \mathcal{L}_{ADSP}(\tilde{\beta}, \theta) := \mathcal{L}(\tilde{\beta}, \theta) + \lambda F(\tilde{\beta}, \theta) \right\}, \tag{15}$$

where $\lambda \geqslant 0$ is the regularization strength. $\mathcal{L}$ and F are the base loss and the FLV defined in (5) and (14), respectively. The parameters $\tilde{\beta}$ and θ can be jointly learned in this model by simultaneously minimizing both base loss and regularization terms. This is a plug-and-play defense strategy that does not need to reform the internal structure of the target system, which preserves the underlying physical theories and laws to the greatest extent possible (see Figure 2).

3.3 Calculation of Functional Local Variation

To implement the proposed ADSP, the FLV $F(\tilde{\beta}, \theta)$ defined in (14) remains to be calculated in a tractable and technically-sound way. Since ϕ^a is itself a function of the learnable parameters $\tilde{\beta}$ and θ, $F(\tilde{\beta}, \theta)$ is a functional of ϕ^a, which increases the difficulty in calculation. Worse still, there are two non-differentiable calculating steps in $\partial_{(\tilde{\beta}, \theta)} F(\tilde{\beta}, \theta)$ that do not support the `autograd` technique in mainstream deep learning architectures: 1) F itself is non-differentiable w.r.t. ϕ^a; 2) ϕ^a is non-differentiable w.r.t. $\tilde{\beta}$ since $\tilde{\beta}$ is calculated by a stochastic sampler.

To overcome the above challenges, we employ the Fréchet subdifferential [47] to extend the function of gradient to the non-differentiable situations. Besides, we notice that the backward propagation in mainstream deep learning architectures is a piecewise calculation, which allows us to fix the parameters $(\tilde{\beta}, \theta)$ when differentiating the FLV F w.r.t. ϕ^a, then we can further compute $\partial_{(\tilde{\beta}, \theta)} \phi^a$ (see Figure 1b). This mechanism allows us to compute the functional $F(\tilde{\beta}, \theta)$ in a similar way to a function. To simplify notation, we denote $\mathcal{L}(\phi)$ with $\phi \in \mathbb{R}^A$ being a vector of A features when some $(\tilde{\beta}, \theta)$ is fixed.

Definition 1 (The Fréchet subdifferential)**.** The Fréchet subdifferential $\partial_\phi V(\phi)$ is the set of elements $\psi \in \mathbb{R}^A$ that satisfy the following criterion.

$$\partial_\phi V(\phi) := \left\{ \psi : \liminf_{\substack{\bar{\phi} \to \phi \\ \bar{\phi} \neq \phi}} \frac{V(\bar{\phi})) - V(\phi) - \langle \psi, \bar{\phi} - \phi \rangle}{\|\bar{\phi} - \phi\|_2} \geqslant 0 \right\}. \tag{16}$$

If V is differentiable at ϕ, then $\partial_\phi V(\phi)$ reduces to a standard gradient. (5) indicates that $\mathcal{L}$ is twice-differentiable w.r.t. ϕ. Let $V(\phi) = \|\partial_\phi \mathcal{L}(\phi)\|_2$ in (16), it can be seen that $\|\partial_\phi \mathcal{L}(\phi)\|_2$ is non-differentiable only when $\partial_\phi \mathcal{L}(\phi) = 0$. In this case, $\psi = 0$ satisfies (16) because $\|\partial_\phi \mathcal{L}(\bar{\phi})\|_2 \geqslant \|\partial_\phi \mathcal{L}(\phi)\|_2 = 0$. Therefore, we can directly set $\partial_\phi \|\partial_\phi \mathcal{L}(\phi)\|_2 = 0$ when $\partial_\phi \mathcal{L}(\phi) = 0$. Then we can rewrite the backward module for $\partial_\phi \|\partial_\phi \mathcal{L}(\phi)\|_2$ as follows:

$$\partial_\phi \|\partial_\phi \mathcal{L}(\phi)\|_2 \leftarrow \begin{cases} \frac{(\partial_\phi \mathcal{L}(\phi))^\top \partial_\phi^2 \mathcal{L}(\phi)}{\|\partial_\phi \mathcal{L}(\phi)\|_2} & \text{if } \partial_\phi \mathcal{L}(\phi) \neq 0, \\ 0 & \text{if } \partial_\phi \mathcal{L}(\phi) = 0, \end{cases} \tag{17}$$

which can be directly used as an ordinary gradient in the `autograd` scheme. $\partial_\phi^2 \mathcal{L}(\phi)$ denotes the Hessian (second-order derivative) of $\mathcal{L}$ w.r.t. ϕ. On the other hand, we can directly compute $\partial_\theta \phi$ via the `autograd` scheme. As for $\partial_{\tilde{\beta}} \phi$, we notice that it is a differentiation following an expectation under $\langle \cdot \rangle_{\tilde{H}}$, based on MCMC sampling [21, 43, 22]. By interchanging differentiation and expectation (which is valid in the current case), we can estimate and rewrite $\partial_{\tilde{\beta}} \phi$ by (18), then compute (19) via the `autograd` scheme:

$$\partial_{\tilde{\beta}} \phi = \langle (\sum_{\langle ij \rangle} x'_i x'_j - \langle \sum_{\langle ij \rangle} x'_i x'_j \rangle_{\tilde{H}}) \cdot (\phi(G_\theta(\tilde{x}')) - \phi(x')) \rangle_{\tilde{H}}, \tag{18}$$

$$\partial_{(\tilde{\beta},\theta)} F(\tilde{\beta}, \theta) = \partial_{(\tilde{\beta},\theta)} \phi \cdot \partial_\phi \| \partial_\phi \mathcal{L}(\phi) \|_2. \tag{19}$$

This enables the training of $(\tilde{\beta}, \theta)$ under our defense scheme ADSP (15) (see Figure 1b). The entire attack and defense loop is presented in Appendix A.3.

4 Experiments

In this section, we evaluate the proposed attack and defense framework for retrieving the temperature parameters in a standard 2D Ising model and a 3-state Potts model, which are primary lattice models of interacting magnetic spins. We show that the proposed defense strategy ADSP can retrieve the ground-truth solution with a high accuracy under a small Gaussian adversarial perturbation, under the Projected Gradient Descent (PGD) attack [41], or without perturbation. Moreover, ADSP also shows effective and robust performance in other evaluating metrics, such as loss function, gradient magnitude, Hessian determinant, and feature difference. The experimental settings are provided in Appendix A.4. **Extensive additional experiments on more advanced attacks and defenses, as well as other aspects are provided in Appendix A.6.**

4.1 Experimental Results on 2D Ising Model for Perturbed Inputs

Estimated $\tilde{\beta}$. Four main levels of ground-truth temperatures $\beta \in \{0.2, 0.3, 0.4, 0.44\}$ are used in the estimation task. Note that $\beta = 0.44$ is close to the phase transition point where the physical structure is unstable, which is a challenging case for the ML systems. The experimental significance of observing the estimated temperature $\tilde{\beta}$ lies in its role as a dynamically learned, interpretable physical quantity. The entire learning process of $\tilde{\beta}$ via the original system and the proposed ADSP method are plotted in Figure A2, which provides a direct and quantitative validation. The solid lines represent the instantaneous $\tilde{\beta}_t$ along with the learning processes, while the dashed lines represent the mean $\tilde{\beta}$ of the last 100 contrastive divergence steps, which can be used to check the convergence of learning processes. The original method deviates significantly from the three solutions $\tilde{\beta}^* = \beta$, $\tilde{\beta}^* = -\beta$, and $\sinh(2\tilde{\beta}^*)\sinh(2\beta) = 1$, in all the cases. In contrast, ADSP converges smoothly to estimations $\tilde{\beta}$ that are very close to the ground-truth β in all the cases, within 5% relative error. Specifically, ADSP converges to $\tilde{\beta} = 0.210$ around the 400-th step for $\beta = 0.2$, $\tilde{\beta} = 0.285$ around the 800-th step for $\beta = 0.3$, $\tilde{\beta} = 0.392$ around the 900-th step for $\beta = 0.4$, and $\tilde{\beta} = 0.428$ around the 800-th step for $\beta = 0.44$, respectively. These results show that the original method deteriorates significantly under the small Gaussian adversarial perturbations, while ADSP shows good adversarial robustness and estimates the ground-truth macroparameter with acceptable accuracy.

Loss Function. The loss function is a core metric for evaluating the statistical similarity between the learned $\tilde{\beta}$ and the ground-truth β. To dig into the reason why there is a large gap between the original method and ADSP in the estimation $\tilde{\beta}$, we also plot their loss functions along with the learning process in Figure A3. $\mathcal{L}$ and $\mathcal{L}_{ADSP}$ denote the base loss of the original method (or ADSP) and the total loss of ADSP, respectively. The difference $(\mathcal{L}_{ADSP} - \mathcal{L}) = \lambda F$ represents the FLV regularization magnitude in ADSP. Observing the changes in $\mathcal{L}$ and $\mathcal{L}_{ADSP}$ over training iterations (contrastive divergence steps) is the primary basis for determining if the optimization process is converging. A continuously decreasing loss curve indicates that $\tilde{\beta}$ is being effectively optimized, leading the model-generated statistical features to gradually approach those of the ground-truth β. Figure A3 shows that the base loss $\mathcal{L}$ of the original method fluctuates significantly and cannot converge in all the cases. Worse still, the values of $\mathcal{L}$ in the late stage of learning even surpass those in the early stage for multiple times. This indicates that the perturbation noise is transmitted to different parts of the system, and even amplified by the backpropagation scheme (see Figure 2), which sabotages the learning process of the original method. In contrast, both $\mathcal{L}$ and $\mathcal{L}_{ADSP}$ of

ADSP generally decrease and converge to stable levels. The rebounds of $\mathcal{L}$ and $\mathcal{L}_{ADSP}$ around the 400-th step in the $\beta = 0.4$ case correspond to the learning path adjustment for $\tilde{\beta}$, which can be verified by the $\beta = 0.4$ case in Figure A2. Moreover, by comparing the difference $(\mathcal{L}_{ADSP} - \mathcal{L})$ in the plots, one can analyze the impact of the FLV regularization on the optimization process and the final solution, and evaluate the trade-off between suppressing the adversarial perturbation and matching the target features. In Figure A3, $(\mathcal{L}_{ADSP} - \mathcal{L})$ does not expand or even shrinks during the learning process, which indicates that the corresponding FLV is constrained within a certain level. In this way, the perturbation of loss caused by the distortion of key physical features is also controlled within a suitable range, which stabilizes the learning process of ADSP.

Gradient Magnitude. In our experimental framework, monitoring the optimization dynamics is important for validating the convergence and stability of the learning process. A key metric for this analysis is the magnitude of the loss gradient: $\|\partial_{\tilde{\beta}}\mathcal{L}\|_2$ and $\|\partial_\theta\mathcal{L}\|_2$ for the original method or $\|\partial_{\tilde{\beta}}\mathcal{L}_{ADSP}\|_2$ and $\|\partial_\theta\mathcal{L}_{ADSP}\|_2$ for the proposed ADSP, which are plotted in Figure A4. Since $\partial_{\tilde{\beta}}\mathcal{L}$ and $\partial_{\tilde{\beta}}\mathcal{L}_{ADSP}$ are estimated gradients by MCMC sampling, they are generally more volatile than the fully-deterministic gradients $\partial_\theta\mathcal{L}$ and $\partial_\theta\mathcal{L}_{ADSP}$. This can be seen by the gaps of plots between the upper line and the lower line in Figure A4. $\|\partial_{\tilde{\beta}}\mathcal{L}_{ADSP}\|_2$ is significantly smaller than $\|\partial_{\tilde{\beta}}\mathcal{L}\|_2$ in most iterations of all the β cases, which indicates that ADSP achieves more stable learning performance than the original method. Besides, $\|\partial_{\tilde{\beta}}\mathcal{L}_{ADSP}\|_2$ has fewer spikes than $\|\partial_{\tilde{\beta}}\mathcal{L}\|_2$, hence ADSP has a smoother learning process. Moreover, $\|\partial_{\tilde{\beta}}\mathcal{L}_{ADSP}\|_2$ steadily decays over the iterations and asymptotically approaches zero, concurrently with $\mathcal{L}_{ADSP}$ reaching a stable and low level. This confirms the robust performance of ADSP against adversarial perturbations. On the other hand, $\|\partial_\theta\mathcal{L}_{ADSP}\|_2$ is also significantly smaller than $\|\partial_\theta\mathcal{L}\|_2$ in most iterations of all the β cases, except for a main fluctuation around the 400-th step in the $\beta = 0.4$ case, which corresponds to the learning path adjustment for $\tilde{\beta}$. This verifies the attack chain explained in Section 3.1 and Figure 2: the adversarial perturbations invade and corrupt $\partial_\theta\mathcal{L}$, but they are suppressed in $\partial_\theta\mathcal{L}_{ADSP}$.

Hessian Determinant. The Hessian determinant of the loss function also provides important diagnostic and analytical insight into both the optimization process and the geometry of the parameter landscape, particularly when training ML systems to capture critical phenomena in statistical physics systems like the 2D Ising model. In our experimental task, it can be represented by $\det(\partial^2_{\tilde{\beta}}\mathcal{L})$ for the original method or $\det(\partial^2_{\tilde{\beta}}\mathcal{L}_{ADSP})$ for the proposed ADSP, respectively. Note that the second (sub)differential of $\mathcal{L}$ or $\mathcal{L}_{ADSP}$ w.r.t. θ does not exist because the feature extracting masks $\{\phi^a\}$ are discrete element-selecting operators on matrices that only have first-order subdifferential [58, 6]. Figure A5 shows that $\det(\partial^2_{\tilde{\beta}}\mathcal{L})$ fluctuates significantly in most β cases, which implies rapid and severe changes in the local curvature of the loss function $\mathcal{L}$. It indicates that the adversarial perturbation invades the Hessian and causes significant volatility, and the optimizer constantly struggles to escape this ill-conditioned state. In contrast, $\det(\partial^2_{\tilde{\beta}}\mathcal{L}_{ADSP})$ remains much more steady in most β cases except for a brief spike around the 400-th step in the $\beta = 0.4$ case, which corresponds to the learning path adjustment for $\tilde{\beta}$. For the $\beta = 0.44$ case, $\det(\partial^2_{\tilde{\beta}}\mathcal{L}_{ADSP})$ has similar fluctuations to those of $\det(\partial^2_{\tilde{\beta}}\mathcal{L})$ due to the unstability of phase transition, but the former is actually less volatile than in the $\beta = 0.4$ case.

Feature Difference. To assess the stability of learning key physical features, we can examine the feature differences between the regenerated part and the ground-truth part: $\{\langle\phi^a(G_\theta(\tilde{x}))\rangle_{\tilde{H}} - \langle\phi^a(x)\rangle_H\}_{a=1}^{13}$. Note that it is hardly possible to retrieve zero feature differences due to the different random dynamics between $\tilde{x}$ and x in MCMC sampling. Hence this experiment mainly evaluates the stability of feature differences. Figure A6 shows the results of the original method and the proposed ADSP. The original method experiences violent fluctuations in most β cases, which implies that its architecture has limited inherent expressiveness relative to the complexity of the target feature space under adversarial perturbations. More importantly, such fluctuations indicate that the gradients of original method in learning are noisy, which leads to erratic and volatile movements in the parameter space. In contrast, ADSP achieves stable differences in most β cases, which indicates that it provides a robust gradient estimate that can suppress the adversarial noise. Hence ADSP keeps each parameter update highly representative of the overall distribution matching objective, which leads to more consistent and predictable learning improvement. Moreover, the small, residual fluctuations also indicate that the learned distribution is stable and accurately mimics the thermodynamic properties of

the ground-truth physical system. For the $\beta = 0.44$ case, ADSP seems a little more volatile than the original method, but this exactly reflects the unstable physical structure of phase transition.

4.2 Experimental Results on 2D Ising Model for Clean Inputs

Experimental results of the original method and the proposed ADSP for the estimated $\tilde{\beta}$, loss function, gradient magnitude, Hessian determinant, and feature difference with clean inputs are shown in Figures A7, A8, A9, A10, and A11, respectively. Compared with the perturbed input scenario, the original method obtains more accurate estimated $\tilde{\beta}$ and less fluctuated evaluating metrics in the clean input scenario. However, some results are still unsatisfactory (e.g., the $\beta \in \{0.3, 0.44\}$ cases), which show the unstable performance of the original method. In contrast, ADSP not only obtains accurate estimated $\tilde{\beta}$ within 5% relative error, but also achieves less fluctuated evaluating metrics in all the β cases, which are similar to the results for the perturbed input scenario. In summary, ADSP retrieves the ground-truth macroparameter with a high accuracy no matter whether the adversarial perturbation is present or absent.

4.3 Experimental Results on 2D Ising model for Perturbed Features

Besides the inputs x and $\tilde{x}$, we also examine another attack surface: the key physical features $\{\phi^a(x)\}_{a=1}^{13}$ and $\{\phi^a(G_\theta(\tilde{x}))\}_{a=1}^{13}$. These features are also vulnerable because: 1) $\phi^a \in \mathbb{R}$, which is a relaxed physical feasibility constraint where injected perturbations are inconspicuous; 2) $\{\phi^a\}_{a=1}^{13}$ are higher-level features with stronger physical semantics than x, and thus have a greater impact on the target system performance. In this experiment, the inputs x and $\tilde{x}$ are clean but the key features are perturbed: $\phi'^a = \phi^a + \delta$ and $\tilde{\phi}'^a = \tilde{\phi}^a + \tilde{\delta}$. Experimental results of the original method and the proposed ADSP for the estimated $\tilde{\beta}$, loss function, gradient magnitudes, Hessian determinants, and feature differences are shown in Figures A12, A13, A14, A15, and A16, respectively. Results are similar to the above two scenarios and ADSP achieves effective and robust performance compared with the original method.

4.4 Experimental Results on 3-state Potts Model for Perturbed Inputs

We conduct experiments on the 3-state Potts model [46], which is compatible with the attack and defense settings in this paper and has a similar physics parameter β. We simplify some overly abstruse details without loss of generality to improve readability. Consider a $d \times d$ periodic square lattice where spins are given by $x_i, x_j \in \{0, 1, 2\}$ (corresponding to the 3 states). The system Hamiltonian and associated Boltzmann probability distribution $p_\beta(x)$ are defined as:

$$H_{\text{Potts}}[x] = -\sum_{\langle ij \rangle} \delta^{Kr}(x_i, x_j), \quad p_\beta(x) = \frac{1}{Z} e^{-\beta H_{\text{Potts}}[x]} = \frac{1}{Z} \exp\left(\beta \sum_{\langle ij \rangle} \delta^{Kr}(x_i, x_j)\right). \tag{20}$$

$\delta^{Kr}(\cdot, \cdot)$ denotes the Kronecker delta, $Z = \sum_x \exp(-\beta H_{\text{Potts}}[x])$, and $p_\beta(x)$ is the probability of x under this Hamiltonian. These symbol conventions are consistent with the 2D Ising model: a larger β corresponds to a more ordered state. Spins are embedded into $\mathbb{Z}_3$ phases $z_i = e^{2\pi \mathrm{i} x_i/3}$ and a real-valued variable for each edge is defined as follows:

$$O_{ij} = \text{Re}(z_i \bar{z}_j) = \cos \frac{2\pi(x_i - x_j)}{3}, \tag{21}$$

where O_{ij} takes only two values: $+1$ for matching spins ($x_i = x_j$) and $-\frac{1}{2}$ for mismatched spins ($x_i \neq x_j$). As a result, O_{ij} and $\delta^{Kr}(x_i, x_j)$ maintain an exact affine linear relation:

$$O_{ij} = \frac{3}{2}\delta^{Kr}(x_i, x_j) - \frac{1}{2} \quad \Longleftrightarrow \quad \delta^{Kr}(x_i, x_j) = \frac{2}{3}O_{ij} + \frac{1}{3}. \tag{22}$$

The transformed observable Hamiltonian $H_O[\beta, x]$ satisfies:

$$H_O[x] = -\sum_{\langle ij \rangle} O_{ij} = -\frac{3}{2}\sum_{\langle ij \rangle} \delta^{Kr}(x_i, x_j) + \frac{1}{2}N_{edge} = \frac{3}{2}H_{\text{Potts}}[x] + d^2, \tag{23}$$

where $N_{edge} = 2d^2$ denotes the total number of edges.

As shown in Table 1, standard adversarial training (AT) fails to recover the underlying inverse temperature β under PGD attack, which shows severe estimation drift across all tested cases (e.g., overestimating $\beta = 0.2000$ to 0.5706 and $\beta = 0.3000$ to 1.0335 at step 1000). This failure stems from the high sensitivity of empirical risk minimization when gradient perturbations distort high-order statistical features. In contrast, ADSP effectively suppresses adversarial noise propagation via FLV regularization. Across all target $\beta \in \{0.2000, 0.3000, 0.4000, 0.4400\}$, ADSP converges toward the ground-truth values and achieves accurate estimations of 0.1948, 0.2887, 0.3832, and 0.4203 at step 1000 (within 5% relative error). Hence ADSP keeps good adversarial robustness and thermodynamic consistency across multi-state spin models.

Table 1: Defense comparison between ADSP and AT under the PGD attack on 3-state Potts model.

	CD Step					
β	0	200	400	600	800	1000
PGD+ADSP						
0.2000	0.1100	0.1220	0.1401	0.1582	0.1764	**0.1948**
0.3000	0.1100	0.1401	0.1764	0.2133	0.2506	**0.2887**
0.4000	0.1100	0.1552	0.2116	0.2645	0.3250	**0.3832**
0.4400	0.1100	0.1603	0.2220	0.2810	0.3487	**0.4203**
PGD+AT						
0.2000	0.1100	0.5885	0.5517	0.5762	0.5677	0.5706
0.3000	0.1100	1.0301	1.0106	1.0369	1.0337	1.0335
0.4000	0.1100	0.7515	0.7260	0.7154	0.7115	0.7102
0.4400	0.1100	0.6763	0.6478	0.6308	0.6251	0.6231

5 Conclusion and Discussion

This work introduces a novel threat model and defense methodology for machine learning (ML) systems employed in statistical physics. It identifies a critical vulnerability: ML models trained on data from physical systems are susceptible to small, physically plausible perturbations that are concealed by the statistical nature of the underlying physics. The proposed threat model indicates that low-magnitude, i.i.d. Gaussian noise can be injected into the input of the target system by human or natural, deliberate or unconscious activities. The core insight is that while this noise is designed to average to zero and thus be concealed in macroscopic observables (which are statistical expectations over many samples), it successfully sabotages the ML pipeline. We demonstrate how these small perturbations are amplified during the learning process, interfering with network components like attention mechanisms and corrupting gradient calculations, ultimately leading to the failure of the target system to recover correct physical parameters.

As a countermeasure, we propose the Adversarial Defense for Statistical Physics (ADSP) framework, which is equipped with a Functional Local Variation (FLV) regularizer. The FLV quantifies the sensitivity of the base loss to perturbations in a set of key physical features. By penalizing this variation, ADSP aims to suppress the random fluctuations induced by the adversarial noise, and stabilize the training process. Since the FLV term is non-differentiable, we provide a subdifferential calculation scheme, which makes ADSP compatible with mainstream deep learning architectures. Experiments on a representative task of recovering the temperature parameters in a 2D Ising model and a 3-state Potts model show that the original target ML system converges to incorrect physical parameters under the stealthy noise attack. In contrast, ADSP successfully suppresses the noise and retrieves the ground-truth solution with a high accuracy, showing robust performance with stable losses, gradient magnitudes, Hessian determinants, and feature difference metrics.

In future work, the recovery accuracy of the defense strategy can be further improved by designing more inherently robust and complex ML systems. Besides, stronger threat models with more advanced white-box attacks can also be investigated. Moreover, the impact of the above vulnerabilities on higher-stakes scientific domains, such as pharmaceutical engineering, climate modeling, and particle physics, can be further assessed.

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Code is available at `https://github.com/laizhr/ADSP`.

References

[1] Maksym Andriushchenko, Francesco Croce, Nicolas Flammarion, and Matthias Hein. Square attack: A query-efficient black-box adversarial attack via random search. In *Computer Vision – ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXIII*, page 484–501, Berlin, Heidelberg, 2020. Springer-Verlag.

[2] Martin Arjovsky, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz. Invariant risk minimization, 2020.

[3] Pierre Baldi, Peter Sadowski, and Daniel Whiteson. Searching for exotic particles in high-energy physics with deep learning. *Nature communications*, 5(1):4308, 2014.

[4] Jörg Behler and Michele Parrinello. Generalized neural-network representation of high-dimensional potential-energy surfaces. *Phys. Rev. Lett.*, 98:146401, Apr 2007.

[5] Suresh Bishnoi, Jayadeva Jayadeva, Sayan Ranu, and NM Anoop Krishnan. Brognet: Momentum-conserving graph neural stochastic differential equation for learning brownian dynamics. In *The Twelfth International Conference on Learning Representations*, 2024.

[6] Jian-Feng Cai, Emmanuel J. Candès, and Zuowei Shen. A singular value thresholding algorithm for matrix completion. *SIAM Journal on Optimization*, 20(4):1956–1982, 2010.

[7] Gustau Camps-Valls, Miguel-Ángel Fernández-Torres, Kai-Hendrik Cohrs, Adrian Höhl, Andrea Castelletti, Aytac Pacal, Claire Robin, Francesco Martinuzzi, Ioannis Papoutsis, Ioannis Prapas, et al. Artificial intelligence for modeling and understanding extreme weather and climate events. *Nature Communications*, 16(1):1919, 2025.

[8] Nicholas Carlini, Pratyush Mishra, Tavish Vaidya, Yuankai Zhang, Micah Sherr, Clay Shields, David Wagner, and Wenchao Zhou. Hidden voice commands. In *Proceedings of the 25th USENIX Conference on Security Symposium*, SEC'16, page 513–530, USA, 2016. USENIX Association.

[9] Nicholas Carlini and David Wagner. Towards evaluating the robustness of neural networks. In *2017 IEEE Symposium on Security and Privacy (SP)*, pages 39–57. IEEE, 2017.

[10] T. Chan, S. Esedoglu, F. Park, and A. Yip. *Total Variation Image Restoration: Overview and Recent Developments*, pages 17–31. Springer US, Boston, MA, 2006.

[11] T.F. Chan and S. Esedoglu. Aspects of total variation regularized l^1 function approximation. *SIAM Journal on Applied Mathematics*, (65):1817–1837, 2005.

[12] Terrence Chen, Wotao Yin, Xiang Sean Zhou, Dorin Comaniciu, and Thomas S. Huang. Total variation models for variable lighting face recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 28(9):1519–1524, 2006.

[13] Zhenjia Chen, Zhenyuan Lin, Ji Yang, Cong Chen, Di Liu, Liuting Shan, Yuanyuan Hu, Tailiang Guo, and Huipeng Chen. Cross-layer transmission realized by light-emitting memristor for constructing ultra-deep neural network with transfer learning ability. *Nature Communications*, 15(1):1930, 2024.

[14] Francesco Croce and Matthias Hein. Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks. *Proceedings of the 37th International Conference on Machine Learning (ICML)*, pages 2206–2216, 2020.

[15] Francesco Croce and Matthias Hein. Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks. In Hal Daumé III and Aarti Singh, editors, *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 2206–2216. PMLR, 13–18 Jul 2020.

[16] Felix Dangel, Johannes Müller, and Marius Zeinhofer. Kronecker-factored approximate curvature for physics-informed neural networks. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 34582–34636. Curran Associates, Inc., 2024.

[17] C. Dasgupta and B. I. Halperin. Phase transition in a lattice model of superconductivity. *Phys. Rev. Lett.*, 47:1556–1560, Nov 1981.

[18] Kevin Eykholt, Ivan Evtimov, Earlence Fernandes, Bo Li, Amir Rahmati, Florian Tramèr, Atul Prakash, Tadayoshi Kohno, and Dawn Song. Physical adversarial examples for object detectors. In *Proceedings of the 12th USENIX Conference on Offensive Technologies*, WOOT'18, page 1, USA, 2018. USENIX Association.

[19] Kevin Eykholt, Ivan Evtimov, Earlence Fernandes, Bo Li, Amir Rahmati, Chaowei Xiao, Atul Prakash, Tadayoshi Kohno, and Dawn Song. Robust physical-world attacks on deep learning visual classification. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1625–1634, 2018.

[20] Ian J Goodfellow, Jonathon Shlens, and Christian Szegedy. Explaining and harnessing adversarial examples. In *International Conference on Learning Representations (ICLR)*, 2015.

[21] Evan Greensmith, Peter L Bartlett, and Jonathan Baxter. Variance reduction techniques for gradient estimates in reinforcement learning. *Journal of Machine Learning Research*, 5(Nov):1471–1530, 2004.

[22] Prateek Gupta, Andrea Ferrari, and Nabil Iqbal. A machine learning approach to duality in statistical physics. In *Proceedings of the 42st International Conference on Machine Learning*. PMLR, 2025.

[23] Tomoharu Iwata and Yusuke Tanaka. Symplectic neural gaussian processes for meta-learning hamiltonian dynamics. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence*, IJCAI '24, 2024.

[24] Eric Jang, Shixiang Gu, and Ben Poole. Categorical reparameterization with gumbel-softmax, 2017.

[25] Mehran Kardar. *Statistical Physics of Fields*. Cambridge University Press, 2007.

[26] Yevgeny Kats and Edo Ofir. From strange-quark tagging to fragmentation tagging with machine learning. *Phys. Rev. D*, 111:034003, Feb 2025.

[27] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015.

[28] Dmitrii Kochkov, Jamie A Smith, Ayya Alieva, Qing Wang, Michael P Brenner, and Stephan Hoyer. Machine learning–accelerated computational fluid dynamics. *Proceedings of the National Academy of Sciences*, 118(21):e2101784118, 2021.

[29] Cevahir Koprulu, Franck Djeumou, and ufuk topcu. Neural stochastic differential equations for uncertainty-aware offline RL. In *The Thirteenth International Conference on Learning Representations*, 2025.

[30] H. A. Kramers and G. H. Wannier. Statistics of the two-dimensional ferromagnet. part i. *Phys. Rev.*, 60:252–262, Aug 1941.

[31] Zhao-Rong Lai and Weiwen Wang. Invariant risk minimization is a total variation model. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235, pages 25913–25935. PMLR, 21–27 Jul 2024.

[32] Zhao-Rong Lai, Xiwen Yuan, Ziliang Chen, Liangda Fang, and Yongsen Zheng. A unified total variation framework for membrane potential perturbation dynamic. In *The Fourteenth International Conference on Learning Representations*, 2026.

[33] Zhao-Rong Lai, Xiwen Yuan, and Jian Weng. Adversarial attack and defense for denoising diffusion sampling. In *The 43rd International Conference on Machine Learning*, 2026.

[34] Zhao-Rong Lai, Zheng-Sen Zhou, Liangda Fang, Yongsen Zheng, and Ziliang Chen. ℓ_1 latent distance based continuous-time graph representation. In *The Fourteenth International Conference on Learning Representations*, 2026.

[35] Huiying Li, Shawn Shan, Emily Wenger, Jiayun Zhang, Haitao Zheng, and Ben Y. Zhao. Blacklight: Scalable defense for neural networks against Query-Based Black-Box attacks. In *31st USENIX Security Symposium (USENIX Security 22)*, pages 2117–2134, Boston, MA, August 2022. USENIX Association.

[36] Ruichen Li, Haotian Ye, Du Jiang, Xuelan Wen, Chuwei Wang, Zhe Li, Xiang Li, Di He, Ji Chen, Weiluo Ren, and Liwei Wang. A computational framework for neural network-based variational monte carlo with forward laplacian. *Nature Machine Intelligence*, 6:209–219, 2 2024.

[37] Shaofei Li, Ziqi Zhang, Haomin Jia, Yao Guo, Xiangqun Chen, and Ding Li. Query provenance analysis: Efficient and robust defense against query-based black-box attacks. In *2025 IEEE Symposium on Security and Privacy (SP)*, pages 1641–1656, 2025.

[38] Yizun Lin, Zhao-Rong Lai, and Cheng Li. A globally optimal portfolio for m-sparse sharpe ratio maximization. In *Advances in Neural Information Processing Systems*, volume 37, pages 17133–17160, 2024.

[39] Yizun Lin, Yangyu Zhang, Zhao-Rong Lai, and Cheng Li. Autonomous sparse mean-CVaR portfolio optimization. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 30440–30456. PMLR, 21–27 Jul 2024.

[40] Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. Towards deep learning models resistant to adversarial attacks. In *International Conference on Learning Representations (ICLR)*, 2018.

[41] Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. Towards deep learning models resistant to adversarial attacks. In *International Conference on Learning Representations (ICLR)*, 2018.

[42] Nicholas Metropolis, Arianna W. Rosenbluth, Marshall N. Rosenbluth, Augusta H. Teller, and Edward Teller. Equation of state calculations by fast computing machines. *The Journal of Chemical Physics*, 21(6):1087–1092, 06 1953.

[43] Shakir Mohamed, Mihaela Rosca, Michael Figurnov, and Andriy Mnih. Monte carlo gradient estimation in machine learning. *Journal of Machine Learning Research*, 21(132):1–62, 2020.

[44] Spandan Mondal and Luca Mastrolorenzo. Machine learning in high energy physics: a review of heavy-flavor jet tagging at the lhc. *The European Physical Journal Special Topics*, 233(15):2657–2686, 2024.

[45] Raymond Muller, Yanmao Man, Ming Li, Ryan Gerdes, Jonathan Petit, and Z Berkay Celik. {VOGUES}: Validation of object guise using estimated components. In *33rd USENIX Security Symposium (USENIX Security 24)*, pages 6327–6344, 2024.

[46] R. B. Potts. Some generalized order-disorder transformations. *Mathematical Proceedings of the Cambridge Philosophical Society*, 48(1):106–109, 1952.

[47] R Tyrrell Rockafellar and Roger J-B Wets. *Variational Analysis*, volume 317. Springer Science & Business Media, 2009.

[48] Franck Rothen, Samuel Klein, Matthew Leigh, and Tobias Golling. Enhancing generalization in high-energy physics using white-box adversarial attacks. *Phys. Rev. D*, 112:016004, Jul 2025.

[49] Leonid I. Rudin, Stanley Osher, and Emad Fatemi. Nonlinear total variation based noise removal algorithms. *Physica D: Nonlinear Phenomena*, 60(1):259–268, 1992.

[50] Daniel Schwalbe-Koda, Aik Rui Tan, and Rafael Gómez-Bombarelli. Differentiable sampling of molecular geometries with uncertainty-based adversarial attacks. *Nature Communications*, 12(1):5104, 2021.

[51] Mahmood Sharif, Sruti Bhagavatula, Lujo Bauer, and Michael K. Reiter. Accessorize to a crime: Real and stealthy attacks on state-of-the-art face recognition. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, CCS '16, page 1528–1540, New York, NY, USA, 2016. Association for Computing Machinery.

[52] Huiming Sun, Lan Fu, Jinlong Li, Qing Guo, Zibo Meng, Tianyun Zhang, Yuewei Lin, and Hongkai Yu. Defense against adversarial cloud attack on remote sensing salient object detection. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 8345–8354, January 2024.

[53] Christian Szegedy, Wojciech Zaremba, Ilya Sutskever, Joan Bruna, Dumitru Erhan, Ian Goodfellow, and Rob Fergus. Intriguing properties of neural networks, 2014.

[54] Florian Tramer, Nicholas Carlini, Wieland Brendel, and Aleksander Madry. On adaptive attacks to adversarial example defenses. In H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 1633–1645. Curran Associates, Inc., 2020.

[55] Oliver T. Unke, Martin Stöhr, Stefan Ganscha, Thomas Unterthiner, Hartmut Maennel, Sergii Kashubin, Daniel Ahlin, Michael Gastegger, Leonardo Medrano Sandonas, Joshua T. Berryman, Alexandre Tkatchenko, and Klaus-Robert Müller. Biomolecular dynamics with machine-learned quantum-mechanical force fields trained on diverse chemical fragments. *Science Advances*, 10(14):eadn4397, 2024.

[56] Jiyao Wang, Hao Lu, Ange Wang, Xiao Yang, Yingcong Chen, Dengbo He, and Kaishun Wu. Physmle: Generalizable and priors-inclusive multi-task remote physiological measurement. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 47(6):4908–4925, 2025.

[57] Yuanchao Wang, Zhao-Rong Lai, and Tianqi Zhong. Out-of-distribution generalization for total variation based invariant risk minimization. In *The Thirteenth International Conference on Learning Representations*, 2025.

[58] G Alistair Watson. Characterization of the subdifferential of some matrix norms. *Linear Algebra and Its Applications*, 170(1):33–45, 1992.

[59] Hui Wei, Zhixiang Wang, Kewei Zhang, Jiaqi Hou, Yuanwei Liu, Hao Tang, and Zheng Wang. Revisiting adversarial patches for designing camera-agnostic attacks against person detection. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 8047–8064. Curran Associates, Inc., 2024.

[60] Eric Wong, Leslie Rice, and J Zico Kolter. Fast is better than free: Revisiting adversarial training. In *The Eighth International Conference on Learning Representations*, 2020.

[61] Tian Xie and Jeffrey C. Grossman. Crystal graph convolutional neural networks for an accurate and interpretable prediction of material properties. *Phys. Rev. Lett.*, 120:145301, Apr 2018.

[62] Huili Zhang, Si Jiang, Xin Wang, Wengang Zhang, Xianzhi Huang, Xiaolong Ouyang, Yefei Yu, Yanqing Liu, Dong-Ling Deng, and L.-M. Duan. Experimental demonstration of adversarial examples in learning topological phases. *Nature Communications*, 13(1):4993, 2022.

[63] Tianfang Zhang, Qiufan Ji, Zhengkun Ye, Md Mojibur Rahman Redoy Akanda, Ahmed Tanvir Mahdad, Cong Shi, Yan Wang, Nitesh Saxena, and Yingying Chen. Safari: Speech-associated facial authentication for ar/vr settings via robust vibration signatures. In *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security*, CCS '24, page 153–167, New York, NY, USA, 2024. Association for Computing Machinery.

[64] Kai Zhao, Qiyu Kang, Yang Song, Rui She, Sijie Wang, and Wee Peng Tay. Adversarial robustness in graph neural networks: A hamiltonian approach. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 3338–3361. Curran Associates, Inc., 2023.

A Appendix

A.1 Related Work

System Vulnerability. As presented in Section 1, several vulnerable parts have been revealed for ML systems in general physical science, such as the input part, the processing architecture, and the training scheme. For example, shallow machine learning models are not capable of learning complex nonlinear functions of the inputs, even with an exhaustive search through manually constructed nonlinear features [3]. A typical solution is to use deep learning architectures to automatically discover the discriminating power and insights contained in the high-level features directly from the raw data, capturing all of the available classification capability [3, 13]. On the other hand, high-quality, expert-labeled samples for training and evaluating ML systems are difficult to obtain since extreme events in physical science are inherently rare. Moreover, it is difficult to eliminate noise, missing values, and biases in the data. The more challenging part is to make a clear statistical definition of extreme events, which complicates data collection and labeling. Possible solutions include transfer learning, causal representation learning, and uncertainty quantification, which explore the mechanisms that trigger extreme events and improve model stability and trustworthiness [7].

Adversarial Attack. In the field of computer vision and acoustics, adversarial perturbations have been intensively investigated, and various attack and defense methods have been developed [9, 19, 18, 45, 51, 59, 8, 63]. In general, an adversarial attack intends to add a small perturbation δ to a benign example x, creating an adversarial example $x' = x + \delta$ that can yield an incorrect output through an ML system [53]:

$$\max_{\|\delta\|_{\mathfrak{p}} \leqslant \epsilon} \mathcal{L}(x + \delta), \tag{24}$$

where $\|\delta\|_{\mathfrak{p}} \leqslant \epsilon$ constrains the $\ell_{\mathfrak{p}}$ norm of the perturbation δ to be no larger than a preset *budget* $\epsilon > 0$. $\mathcal{L}$ denotes the loss function used in training the parameters of the ML system. Since a standard training process aims to minimize $\mathcal{L}$, the adversarial perturbation δ works toward the opposite direction (maximization), in order to sabotage the training process. Adversarial examples make NNs difficult to deploy in security-critical areas, such as autonomous vehicles, due to the risk of unintended actions [9]. Besides, attackers can create seemingly context-consistent motion (e.g., the phantom changing of lanes by a vehicle) to circumvent the existing object detection defenses (like contextual consistency checks). This can be defended by deliberately selecting component reconstruction algorithms with orthogonal vulnerabilities than those of the object detection algorithms [45]. On the other hand, some internal image signal processing pipelines apply complex processing that can attenuate the adversarial perturbation, which degrades the attack performance and cross-device generalizability [59]. In speech processing, an attacker can easily obtain the voice samples of a victim and use them for speech synthesis or replay attacks to bypass the authentication mechanism. A possible solution is to use conductive facial vibrations, which makes the system inherently spoofing-resistant against both speech synthesis and vibration replay attacks that rely on stealing voice biometrics via the audio channel [63].

Violation of Strict Assumptions. Some vulnerabilities of strict assumptions in ML systems have also been revealed. For example, fixed-length vectors and symmetry-invariant transformations of atom coordinates are required in most ML algorithms, which contradicts the fact that a crystal system can have an arbitrary size and makes it difficult to provide chemical insights [61]. Different data acquisition equipment, subject differences, and environmental changes can lead to domain shift, which violates the strict assumption of stationarity and homogeneity of radiative transfer [56]. Some ML systems assume that the detector can accurately discriminate between different particles, which cannot be satisfied due to the inherent physical difficulty of the task. For example, large hadron collider detectors cannot distinguish between charged hadrons (like charged kaons and charged pions), and high-energy neutral strange hadrons are difficult to resolve. To tackle this problem, the graph attention network and the particle transformer are combined to identify complex correlation patterns among all constituents of the jet [26].

Total Variation. We introduce the standard form of TV [49, 12, 10] in the literature of image processing. Denote $f(u, v)$ as a two-dimensional (2D) image with u and v being the indices for these two dimensions. Then the definition of TV is:

$$\int_{\Omega} \sqrt{\left(\frac{\partial f}{\partial u}\right)^2 + \left(\frac{\partial f}{\partial v}\right)^2} \,\mathrm{d}u \,\mathrm{d}v = \int_{\Omega} \|\nabla f\|_2 \,\mathrm{d}u \,\mathrm{d}v, \tag{25}$$

where Ω denotes the domain (pixels) of interest. An intuitive interpretation of TV is to aggregate the variation of pixel value f along two dimensions over the domain of interest. In general, TV is to be minimized in training to preserve sharp features (e.g., edges) of f, since TV itself contains fine-scale textures that are minor details in human vision. TV can also be used as regularization, for example, in the invariant risk minimization training scheme [2, 31, 57]:

$$\min_{\Phi} \left\{ \int_u \mathcal{L}(u \circ \Phi) \, \mathrm{d}u + \lambda \left(\int_u \|\nabla_u \mathcal{L}(u \circ \Phi)\|_2 \, \mathrm{d}u \right)^2 \right\}, \tag{26}$$

where $\mathcal{L}$ is the task loss, and $\lambda > 0$ is the regularization strength. u and Φ denote the classifier and the feature extractor, respectively. They are composed $(u \circ \Phi)$ to form the inner processing structure of the ML system. The second term of (26) is a squared TV w.r.t. u, which aims to suppress the spurious features. This finding motivates the extension of the TV concept and methodology to adversarial defense.

A.2 Key Insight

Section A.1 has already summarized considerable vulnerabilities and the corresponding defense strategies regarding different kinds of ML systems in general physical science. However, such attack and defense strategies are tailored for specific domains, which lack generality and fail to expose deeper, greater, or broader threats. For example, advanced white-box adversarial attack strategies like FGSM [20], PGD [40], and C&W [9] should be used to construct well-crafted examples to breach an advanced ML system. This requires the full knowledge of the target system, which is not easy to fulfill. Black-box or ensemble attack strategies like Square Attack [1] or AutoAttack [14] requires considerable queries from the ML system, which can be efficiently blocked by query-limiting defenses or stateful defense models [37, 35]. Other specific vulnerabilities such as fixed-length inputs and detector limitations can be fixed by improving specific modules of an ML system, such as using graph convolutional NNs and graph attention networks [61, 26]. Such specific vulnerabilities and reactive defense strategies have not dug into the root causes of security issues for ML systems in physical science.

In this work, we focus on the security issues for a representative category of ML systems: the ML systems in statistical physics. Such systems exploit statistical properties of physics to reveal, realize, and verify physical laws, theories, and even conjectures. Developing such systems not only involves esoteric theoretical physics and statistical knowledge, but also requires immense effort, leaving little energy to consider security issues. Moreover, researchers in this field may have scant or even no knowledge of relevant security practices, which frequently results in a lack of robust security mechanisms. On the other hand, the statistical property is a two-sided coin: useful for verifying physical theories, but also capable of concealing attack artifacts. Furthermore, the strict statistical assumptions themselves yield a significant vulnerability, because they can be easily violated and the corresponding outputs can be unreliable or incorrect. For example, Monte Carlo sampling is a widely-used statistical technique in statistical physics [3, 61, 23, 48, 7, 36, 29, 22, 26]. While it estimates a key physical quantity via the Law of Large Numbers, it also conceals the attack perturbation by the same law. This is an intrinsic vulnerability of statistics that can hardly be avoided. In summary, poor security mechanisms in such ML systems can expose scientific research to significant threats, leading to wrong research directions and a waste of research funding.

To develop an appropriate defense scheme for these ML systems, we notice that they usually need to extract some key physical features from a raw input. These features usually contain important physical structural information, such as traditional energy/coupling terms, local angular correlations, intermediate-range correlations, and local gauge invariant quantity (see Table A1). When the input is adversarially perturbed, these structural features may also be distorted, which leads to the consequent bad performance of the entire ML system. To fix this vulnerability, we develop an FLV w.r.t. these features and use it as regularization in the learning process. This strategy can suppress the perturbation of loss function induced by the distortion of the structural features, which reduces the effect brought by the adversarial input.

Our threat model and defense complement some previous studies [50, 62]. In [62], topological phase classifiers suffer because microscopic experimental noise corrupts local correlation signatures. The FLV regularization of ADSP addresses this issue by enforcing local variation smoothness on topological invariants or Wilson loops ϕ^a, which prevents decision boundary distortion. In [50],

uncertainty-based attacks target atomic pair coordinates. In such continuous settings, FLV can be defined over interatomic distance distributions or pairwise force observables $\{\phi^a\}$.

A.3 The Entire Attack and Defense Loop

The entire attack and defense algorithm for the target ML system on 2D Ising model is shown in Algorithm 1. The main task of this system is to estimate an unknown ground-truth macroparameter β. The first phase should have been preparing, observing and analyzing a set of ground-truth samples $\{x_{(n)}\}$ generated from β in the real-world physical experiment. But now since this stage is mimicked by the ML system, $\{x_{(n)}\}$ are simulated by MCMC with a standard Metropolis algorithm [42]. Whether physically generated or virtually simulated, these ground-truth samples $\{x_{(n)}\}$ are then attacked by i.i.d. Gaussian noises due to human or natural, deliberate or unconscious activities, which result in perturbed samples $\{x'_{(n)}\}$. These perturbed samples $\{x'_{(n)}\}$ are all materials that the target ML system has received, while β is inaccessible to the system. After initializing the learnable parameters $\tilde{\beta}_0$ and θ_0, the target system goes into the second phase that learns the unknown β via an iterative scheme. In each iteration (contrastive divergence step), a new set of samples $\{\tilde{x}_{(n)}\}$ are generated from the current estimated macroparameter $\tilde{\beta}_t$ in the same experimental environment as that of ground-truth samples $\{x_{(n)}\}$. At the same time, $\{\tilde{x}_{(n)}\}$ are attacked by i.i.d. Gaussian noises due to the same reason as the above, and result in perturbed samples $\{\tilde{x}'_{(n)}\}$. Next, both $\{\tilde{x}'_{(n)}\}$ and $\{x'_{(n)}\}$ are used in the adversarial learning of the defense strategy with FLV. The parameters $(\tilde{\beta}_{t+1}, \theta_{t+1})$ are then updated, and the system goes into the next iteration. The same attack-defense dynamic continues, until the system has implemented sufficiently many contrastive divergence steps. Last, the learned macroparameter $\tilde{\beta}$ is obtained and compared with one of the three solutions: $\tilde{\beta}^* = \beta$, $\tilde{\beta}^* = -\beta$, or $\sinh(2\tilde{\beta}^*)\sinh(2\beta) = 1$.

Algorithm 1 Adversarial Attack and Defense for Target ML System on 2D Ising Model

Require: Learning rate η, number of samples N, number of contrastive divergence steps T, attack budget ϵ, regularization strength λ, sampling probability p, unknown ground-truth macroparameter β.

`/* Preliminary: Perturbed Samples from the Unknown Ground-truth` β `*/`

Draw NT ground-truth samples $\{x_{(n)}\}_{n=1}^{NT} \sim p(x|\beta)$.

Attack each link: $x'_{(n),i}x'_{(n),j} = x_{(n),i}x_{(n),j} + \delta_{(n),ij}$ with i.i.d. $\delta_{(n),ij} \sim \mathcal{N}(0, \epsilon^2)$ for any neighboring ij and $n \in [1 : NT]$.

`/* Start Learning` β `*/`

Initialize learnable parameters $\tilde{\beta}_0$ and θ_0.

for each contrastive divergence step $t = 1, 2, ..., T$ **do**

  Draw N estimating samples: $\{\tilde{x}_{(n)}\}_{n=N(t-1)+1}^{Nt} \sim p(\tilde{x}|\tilde{\beta}_t)$.

  Attack each link: $\tilde{x}'_{(n),i}\tilde{x}'_{(n),j} = \tilde{x}_{(n),i}\tilde{x}_{(n),j} + \tilde{\delta}_{(n),ij}$

    with i.i.d. $\tilde{\delta}_{(n),ij} \sim \mathcal{N}(0, \epsilon^2)$ for any neighboring ij and $n \in [N(t-1)+1 : Nt]$.

  Compute the base loss $\mathcal{L}$ in (5) with perturbed samples $\{x'_{(n)}\}_{n=N(t-1)+1}^{Nt}$ and $\{\tilde{x}'_{(n)}\}_{n=N(t-1)+1}^{Nt}$, as well as parameters $\tilde{\beta}_t$ and θ_t.

  `/* Defense Starts */`

  Compute the FLV F in (14).

  Plug F in (15) to compute the total loss $\mathcal{L}_{ADSP}$.

  Use (17), (19), and the `autograd` scheme to compute the update gradient $\partial_{(\tilde{\beta},\theta)}\mathcal{L}_{ADSP}$.

  Update the parameters:

    $(\tilde{\beta}_{t+1}, \theta_{t+1}) = (\tilde{\beta}_t, \theta_t) - \eta\partial_{(\tilde{\beta},\theta)}\mathcal{L}_{ADSP}$.

  `/* Defense Ends */`

end for

Ensure: The learned macroparameter $\tilde{\beta}$.

A.4 Experimental Settings (2D Ising Model)

The system Hamiltonian is defined by nearest-neighbor spin interactions, governed by the ground-truth inverse temperature parameter β. The experimental settings generally follow those from the original paper [22], in order to adhere closely to the underlying physical theories and laws. Five repetitions are implemented to derive the mean results.

Neural Network Architecture. A standard 8×8 ($d = 8$) lattice with periodic boundary conditions is used. The lightweight neural network in (10) is employed to parameterize the mapping function G_θ. It takes a 7-dimensional vector representing the first-order link features (see Figure A1) as input, and uses a Gumbel-Softmax layer for hard attention. This forces the model to select a minimal subset of the input features. The 13 key physical features in Table A1 are used as $\{\phi^a\}_{a=1}^{13}$. The parameters θ_2 and θ_3 are initialized from a uniform distribution $\mathcal{U}(-1, 1)$, and the attention weights θ_1 are initialized from a normal distribution $\mathcal{N}(0, \mathbb{I}_7)$, where $\mathbb{I}_7$ denotes the 7-dimensional identity matrix. The temperature estimation $\tilde{\beta}$ is initialized as 0.1.

Optimization and Training. The training scheme aims to minimize the difference in correlation functions between the ground-truth and estimation models. The loss functions used for the original method and ADSP are $\mathcal{L}$ and $\mathcal{L}_{ADSP}$, defined in (5) and (15), respectively. The Adam optimizer [27] with a learning rate of $\eta = 0.01$ is used. The training process involves simultaneous MCMC sampling from both the ground-truth and estimated Hamiltonians to retrieve the target temperature. To improve convergence and reduce the high variance inherent in MCMC-based gradient estimation, a control variate technique [21, 43] is implemented, which is a popular treatment in MCMC sampling. In our defense framework, it is equivalent to using the estimated gradient in (18). The number of samples and the number of contrastive divergence steps are set to $N = 100$ and $T = 1000$, respectively, resulting in a large total number of $NT = 10^5$ samples from either of the ground-truth β or the estimated $\tilde{\beta}$. The attack budget (the standard deviation of the injected Gaussian noise) is set to a small value $\epsilon = 0.001$. Compared with $x_i = \pm 1$, such a small attack budget makes it even more difficult to detect in macroscopic observables. Specifically, the standard deviation of the link observable defined in (7) will be perturbed by only

$$\frac{\epsilon}{\sqrt{2d^2N}} = \frac{0.001}{8 \cdot \sqrt{2 \cdot 100}} \approx 8.8388 \times 10^{-6}. \tag{27}$$

The regularization strength for FLV is set to a small trivial value $\lambda = 0.1$.

A.5 More Tables and Figures (2D Ising Model)

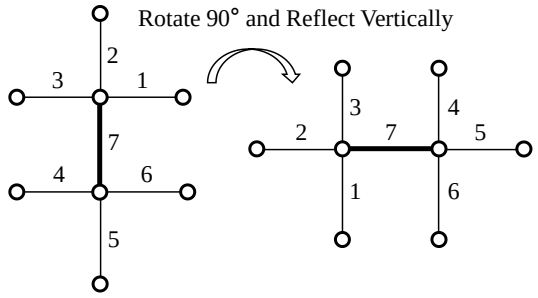


Figure A1: For a given link #7, the 6 neighboring links $\#1 \sim \#6$ and itself constitute the 7 local first-order link features. The NN G_θ is designed to mimic a physical transformation, such as rotating 90° followed by reflecting vertically, which transforms the left local structure to the right one. It can neither check physical feasibility constraints nor restore perturbed samples to clean ones.

Table A1: The 13 local spin-product features for the 2D Ising model.

Feature Index	Geometric Structure	Description (Spin Product)
1	Horizontal Link	Product of two adjacent horizontal spins.
2	Vertical Link	Product of two adjacent vertical spins.
3	L-shape (Type 1)	Product of three spins forming an L-shape (e.g., in a 2×2 corner).
4	L-shape (Type 2)	Another rotational orientation of the L-shape product of three spins.
5	L-shape (Type 3)	Another rotational orientation of the L-shape product of three spins.
6	L-shape (Type 4)	Another rotational orientation of the L-shape product of three spins.
7	Parallel Horizontal	Product of four spins in two adjacent horizontal pairs (horizontal dimer pair).
8	Parallel Vertical	Product of four spins in two adjacent vertical pairs (vertical dimer pair).
9	U-shape (Type 1)	Product of five spins forming an open structure (e.g., missing one link from a 2×3 area).
10	U-shape (Type 2)	Another rotational orientation of the U-shape product.
11	U-shape (Type 3)	Another rotational orientation of the U-shape product.
12	U-shape (Type 4)	Another rotational orientation of the U-shape product.
13	Plaquette (Square)	Product of the four spins forming a single square (smallest loop): $x_i x_j x_k x_l$.

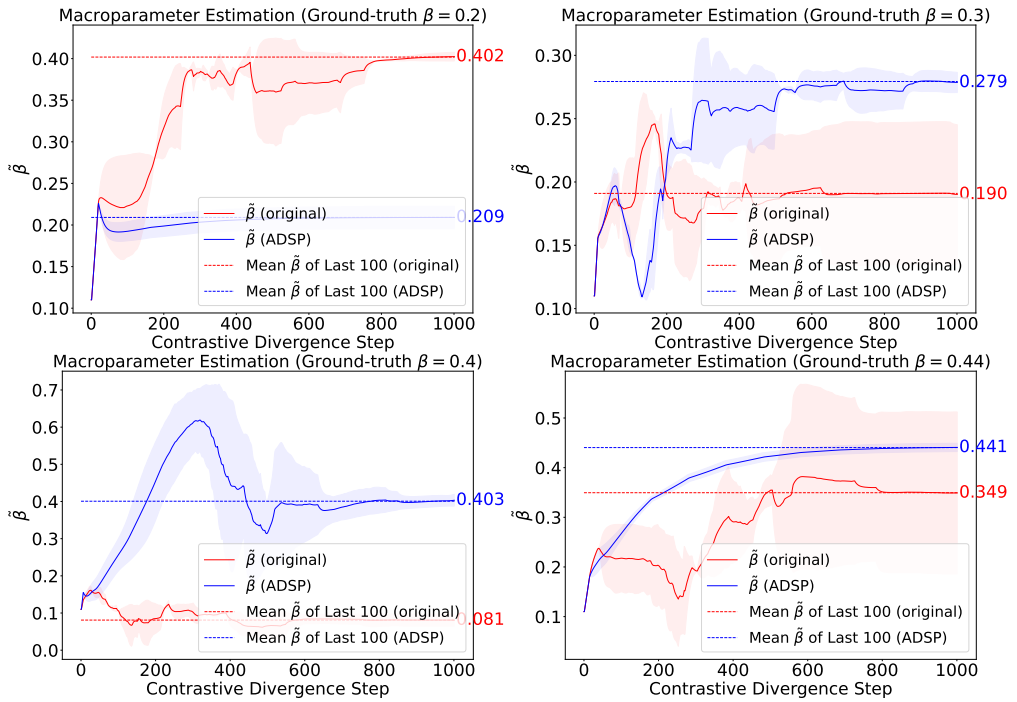


Figure A2: Learning processes of $\tilde{\beta}$ to the ground-truth β via the original method and ADSP with perturbed inputs under the Gaussian noise.

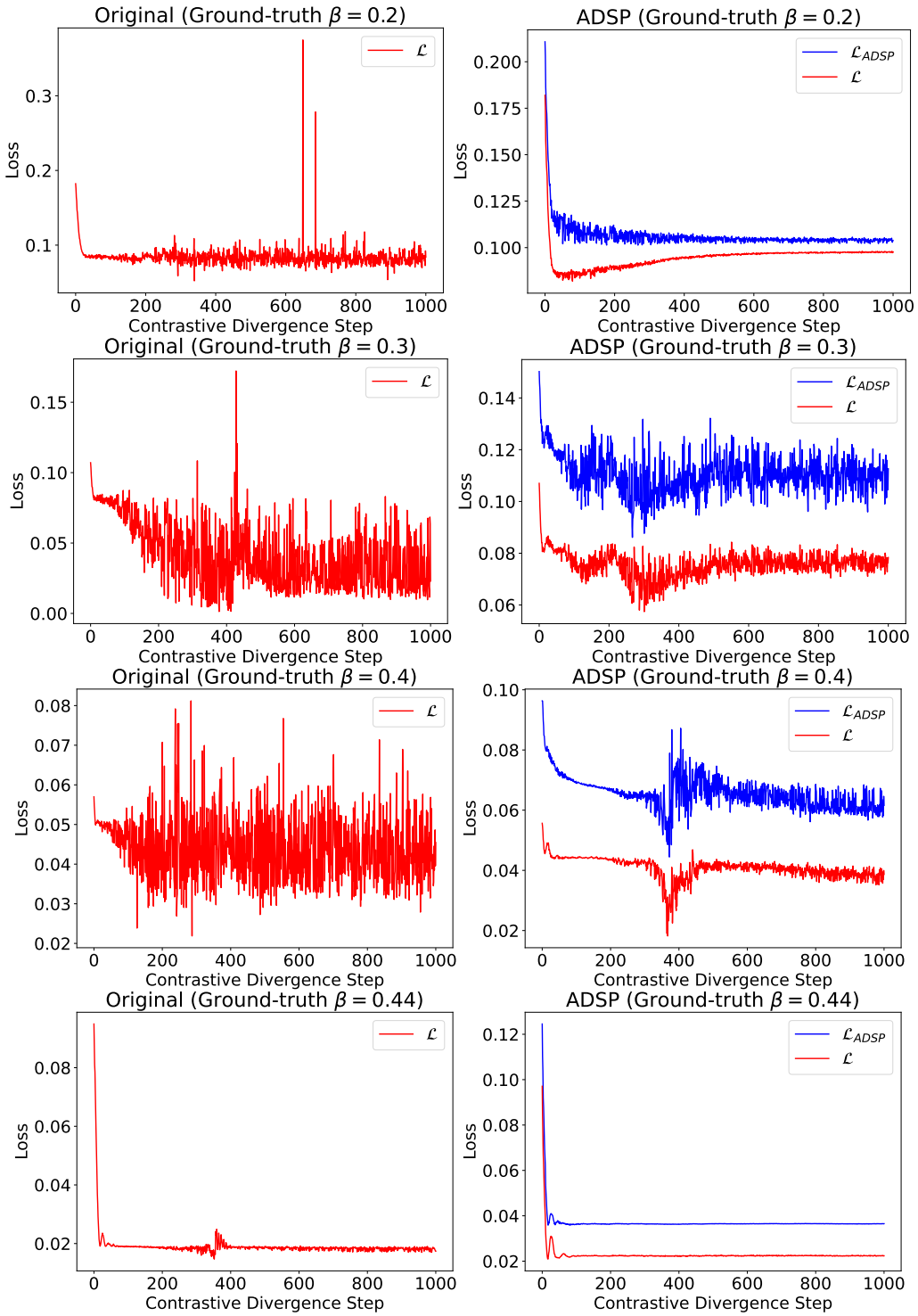


Figure A3: Loss function values of the original method (left) and ADSP (right) during the learning process with perturbed inputs under the Gaussian noise.

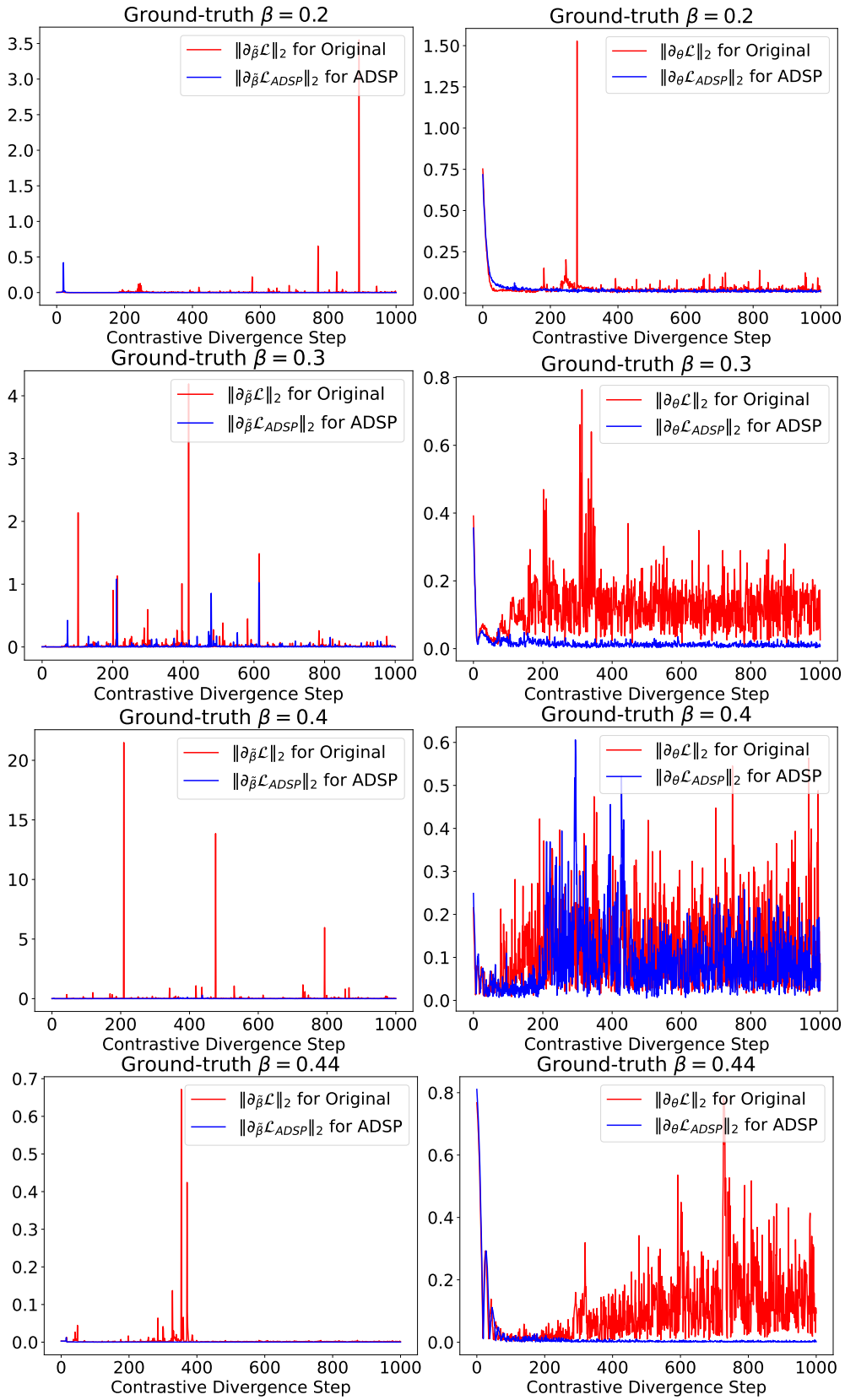


Figure A4: Gradient magnitudes of the original method and ADSP during the learning process with perturbed inputs under the Gaussian noise (left for $\partial_{\tilde{\beta}}$ and right for ∂_{θ}).

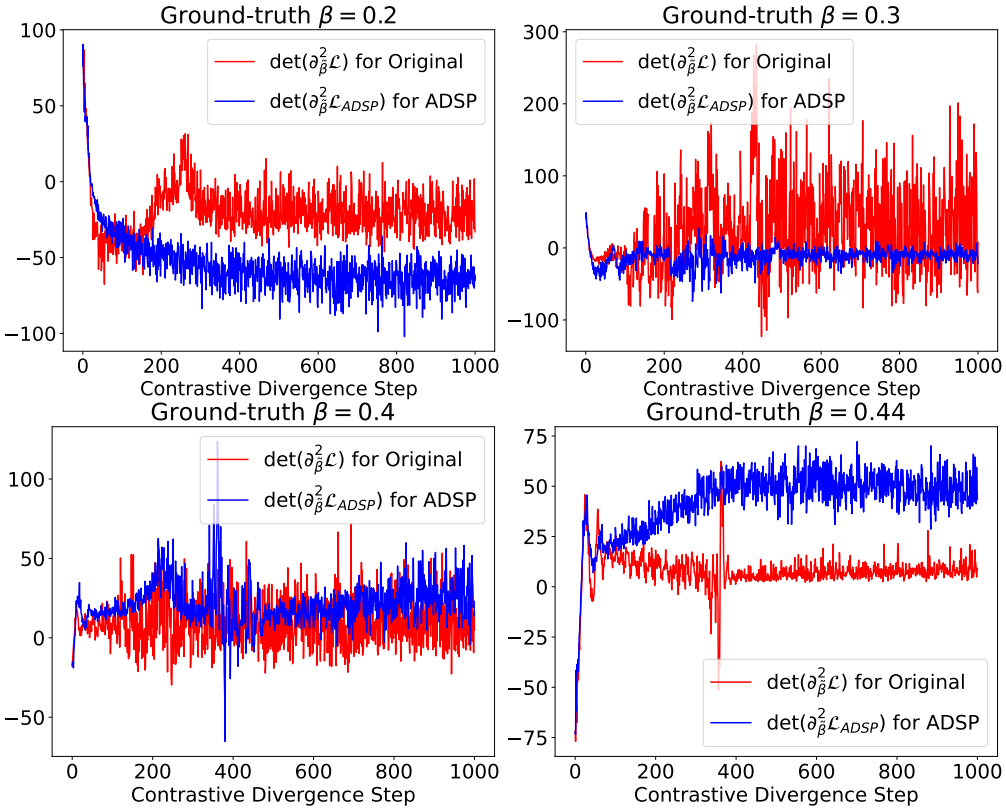


Figure A5: Hessian determinants of the original method and ADSP during the learning process with perturbed inputs under the Gaussian noise.

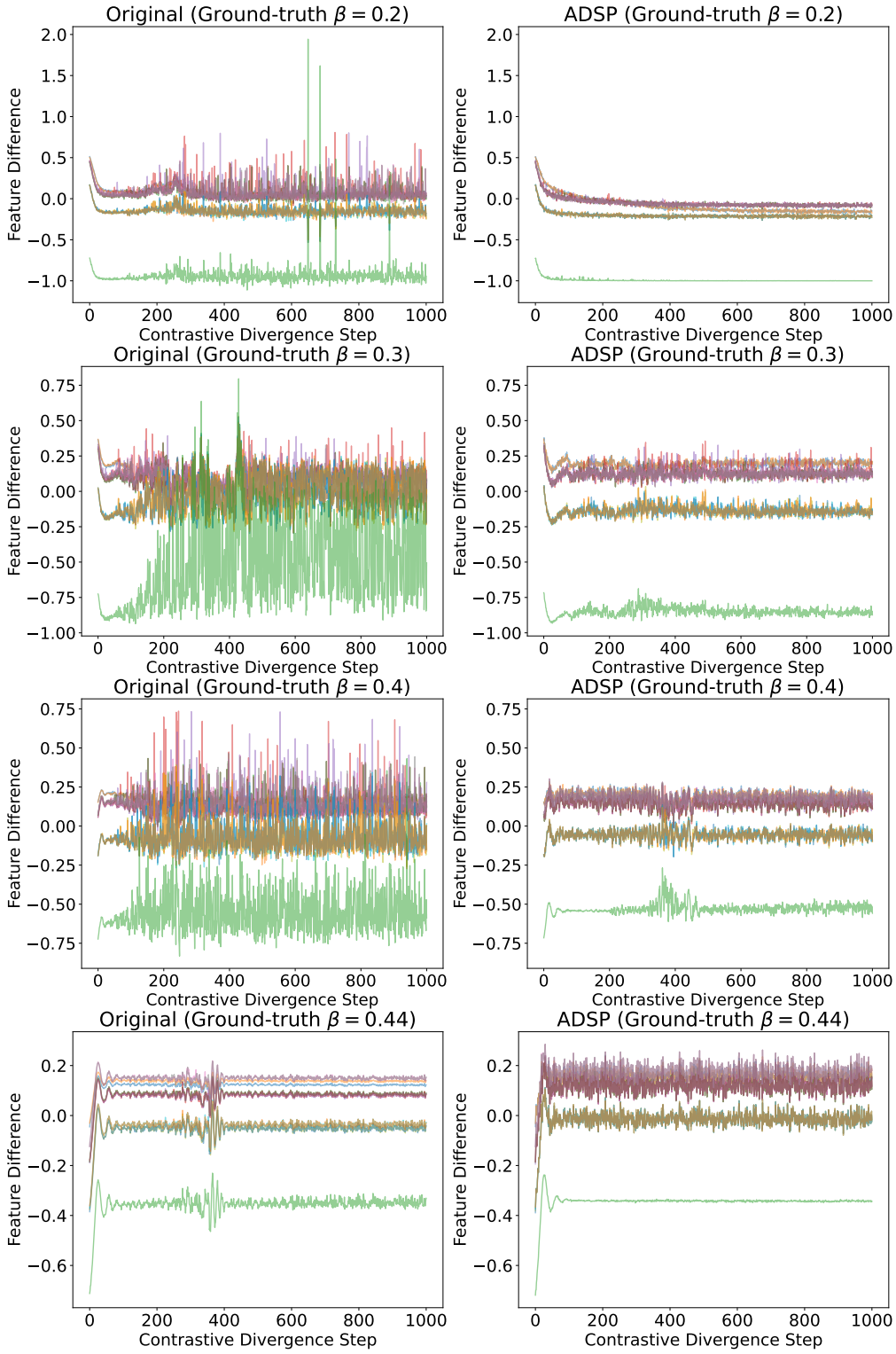


Figure A6: Feature differences $\{\langle \phi^a(G_\theta(\tilde{x}))\rangle_{\tilde{H}} - \langle \phi^a(x)\rangle_H\}_{a=1}^{13}$ of the original method (left) and ADSP (right) during the learning process with perturbed inputs under the Gaussian noise.

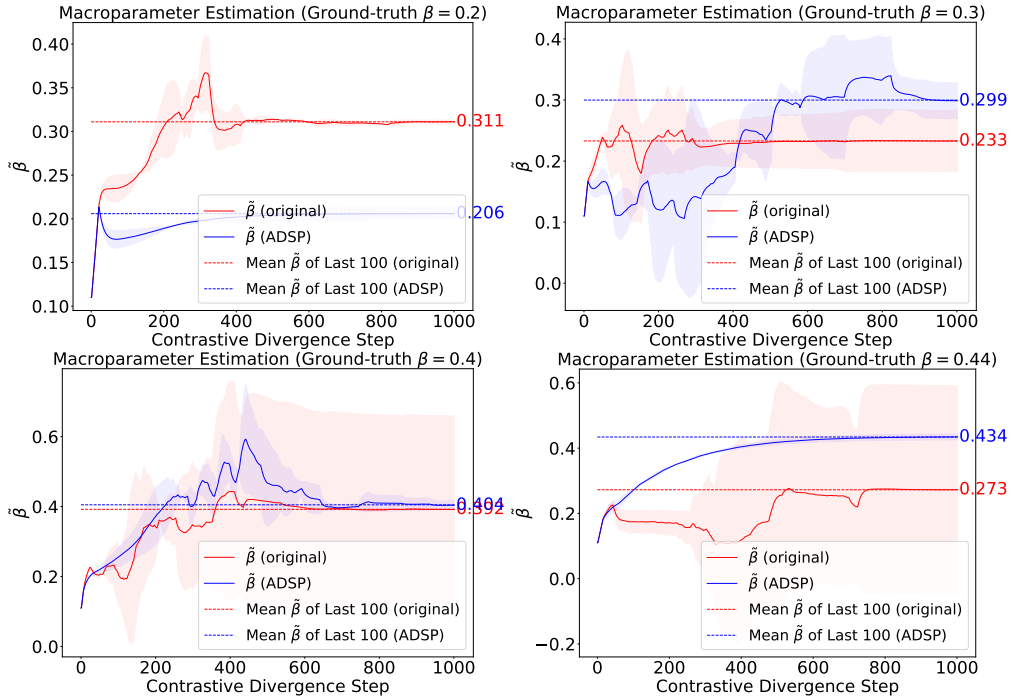


Figure A7: Learning processes of $\tilde{\beta}$ to the ground-truth β via the original method and ADSP with clean inputs.

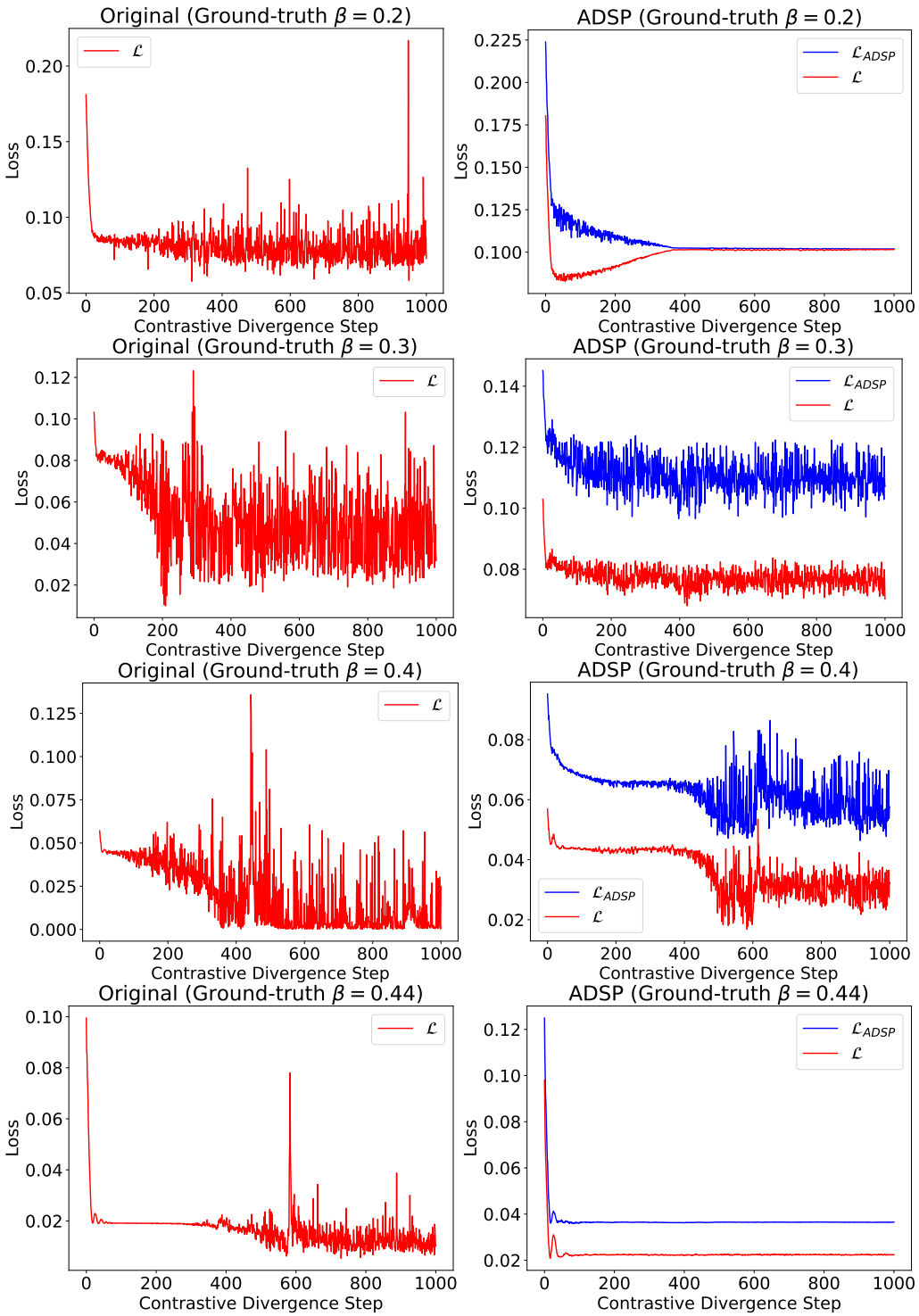


Figure A8: Loss function values of the original method (left) and ADSP (right) during the learning process with clean inputs.

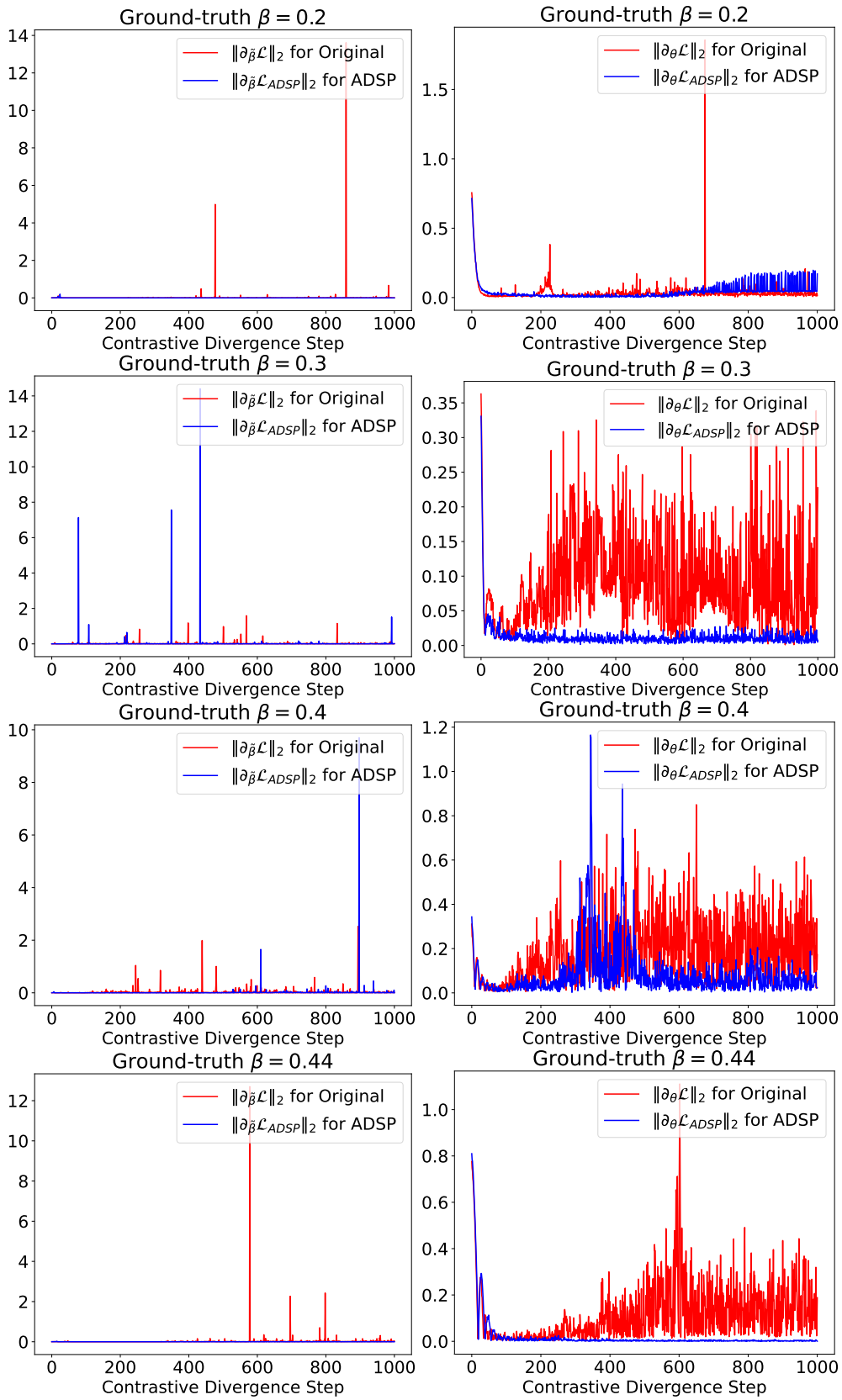


Figure A9: Gradient magnitudes of the original method and ADSP during the learning process with clean inputs (Left for $\partial_{\tilde{\beta}}$ and Right for ∂_{θ}).

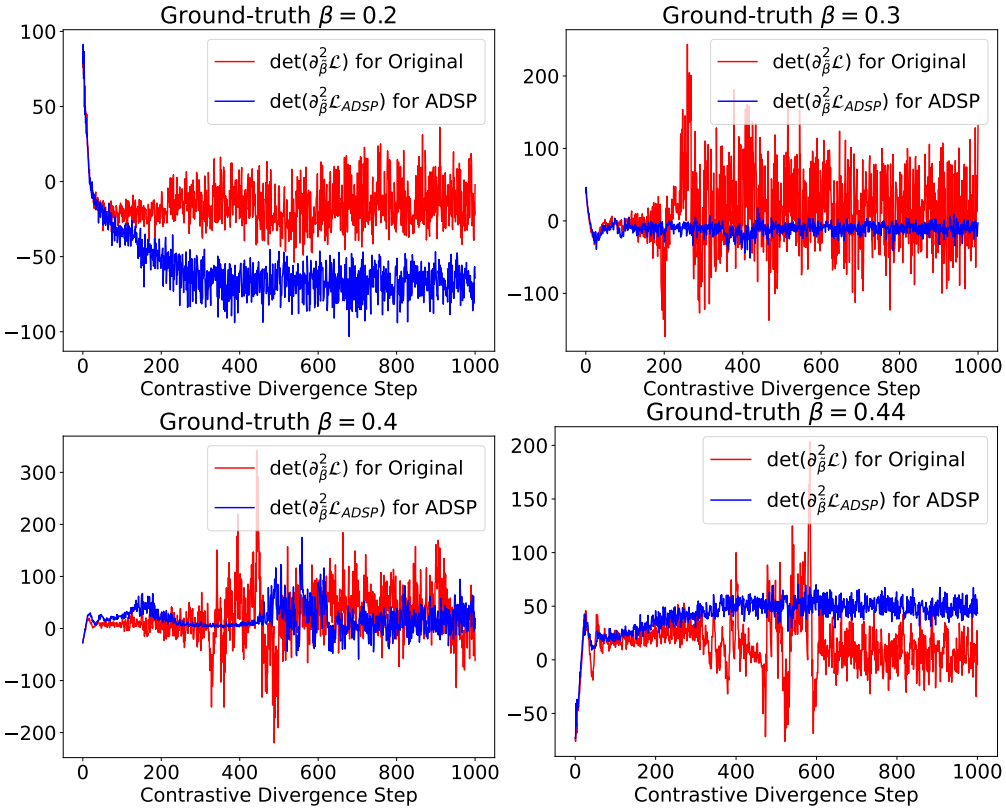


Figure A10: Hessian determinants of the original method and ADSP during the learning process with clean inputs.

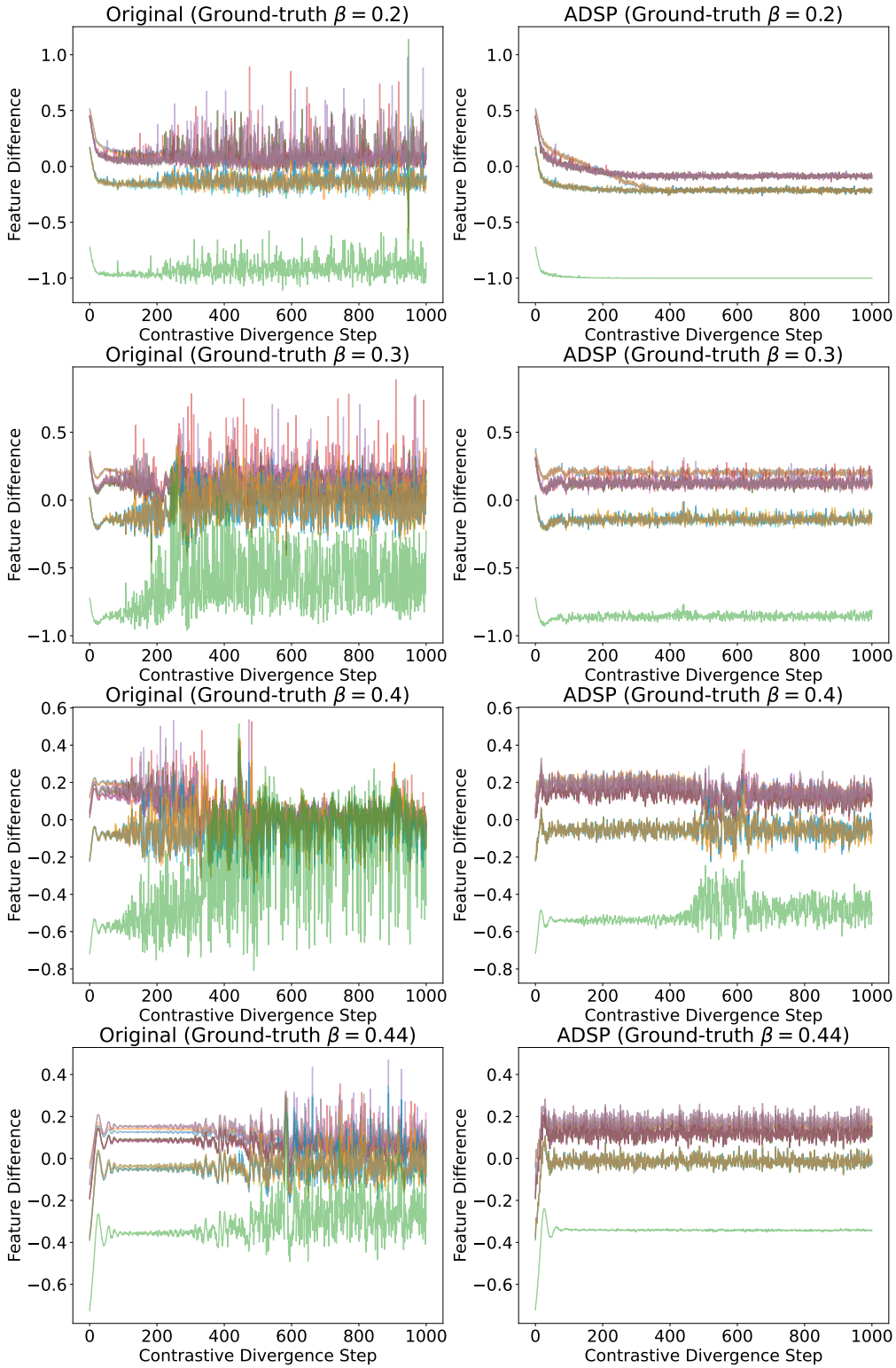


Figure A11: Feature differences $\{\langle \phi^a(G_\theta(\tilde{x}))\rangle_{\tilde{H}} - \langle \phi^a(x)\rangle_H\}_{a=1}^{13}$ of the original method (left) and ADSP (right) during the learning process with clean inputs.

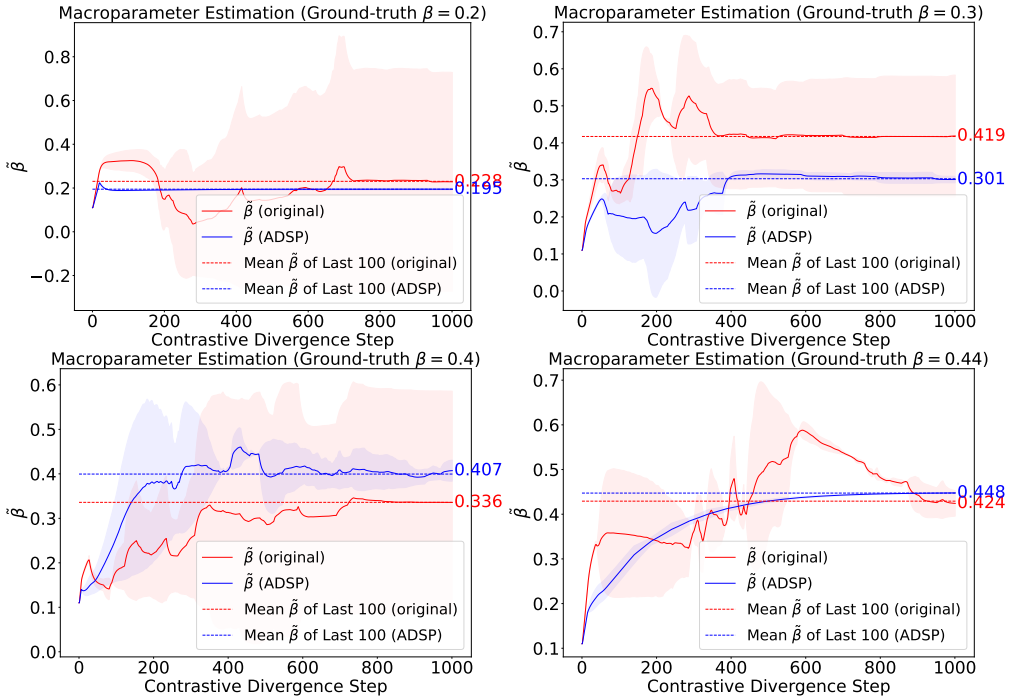


Figure A12: Learning processes of $\tilde{\beta}$ to the ground-truth β of the original method and ADSP during the learning process with perturbed features under the Gaussian noise.

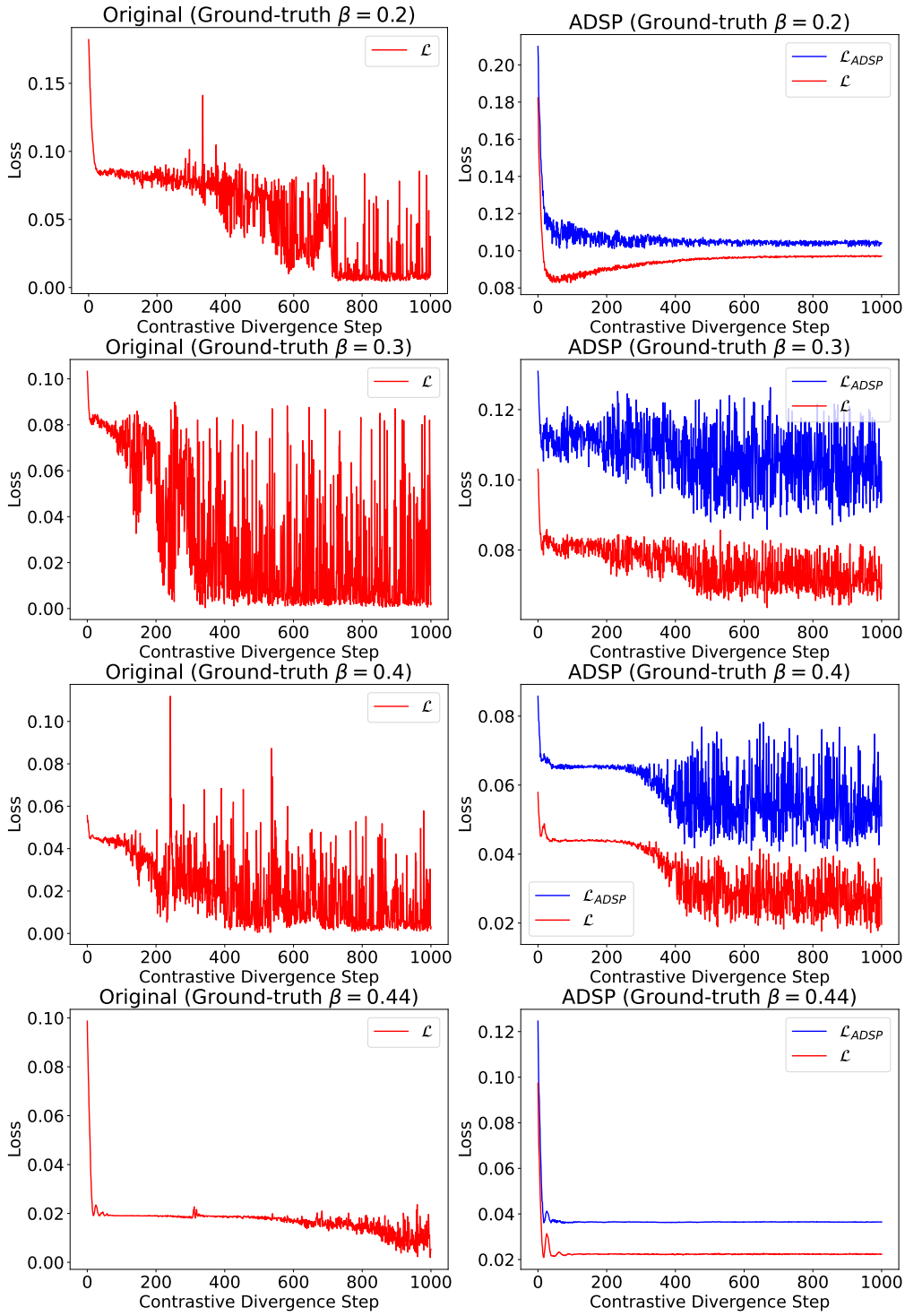


Figure A13: Loss function values of the original method (left) and ADSP (right) during the learning process with perturbed features under the Gaussian noise.

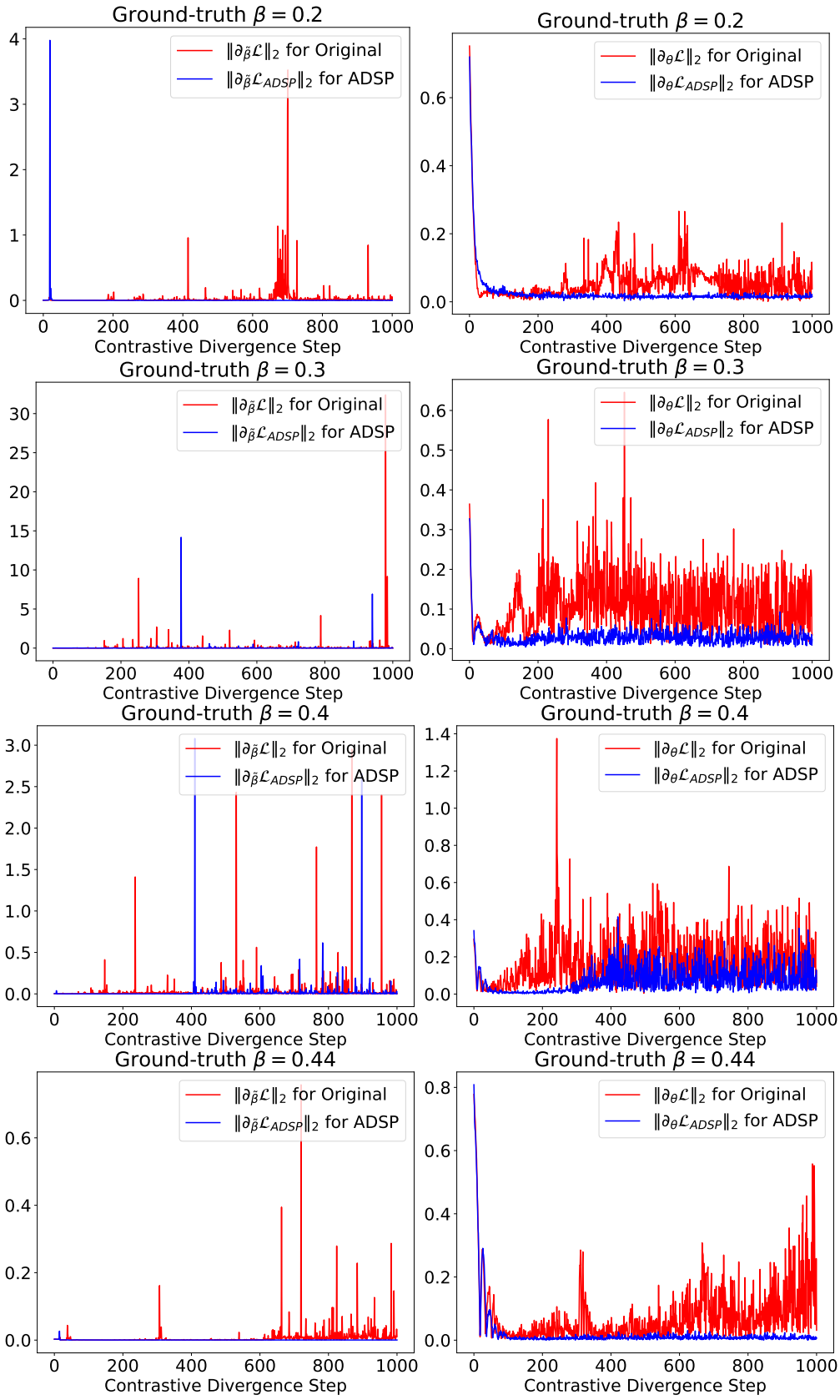


Figure A14: Gradient magnitudes of the original method and ADSP during the learning process with perturbed features under the Gaussian noise (left for $\partial_{\tilde{\beta}}$ and right for ∂_{θ}).

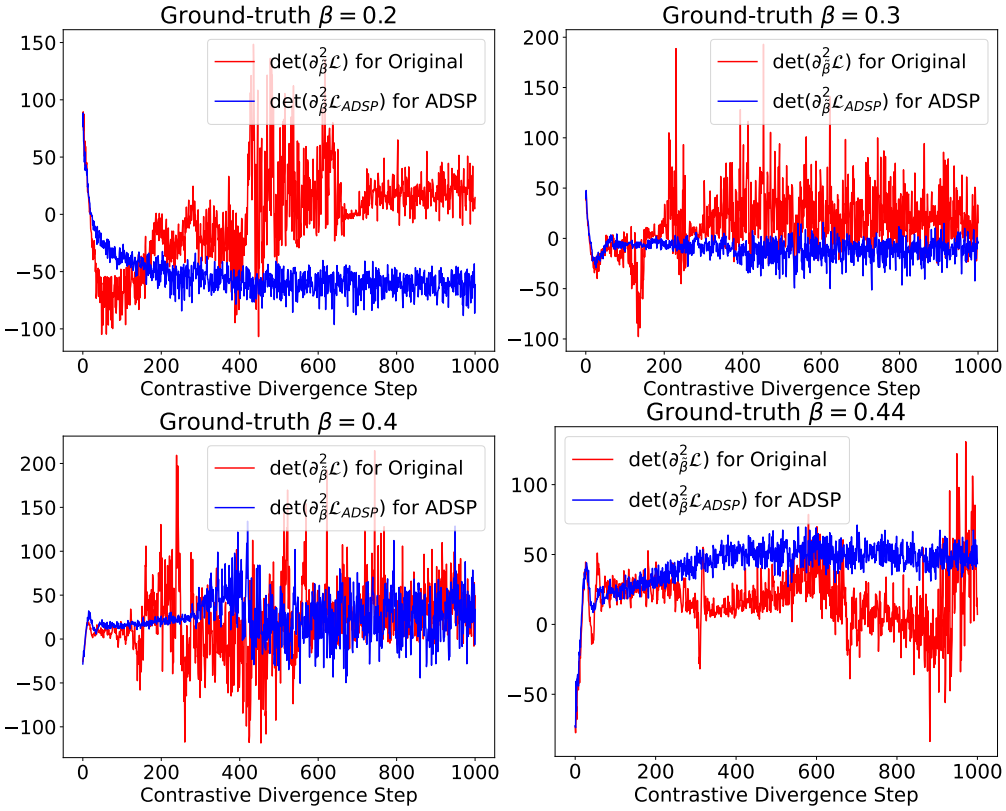


Figure A15: Hessian determinants of the original method and ADSP during the learning process with perturbed features under the Gaussian noise.

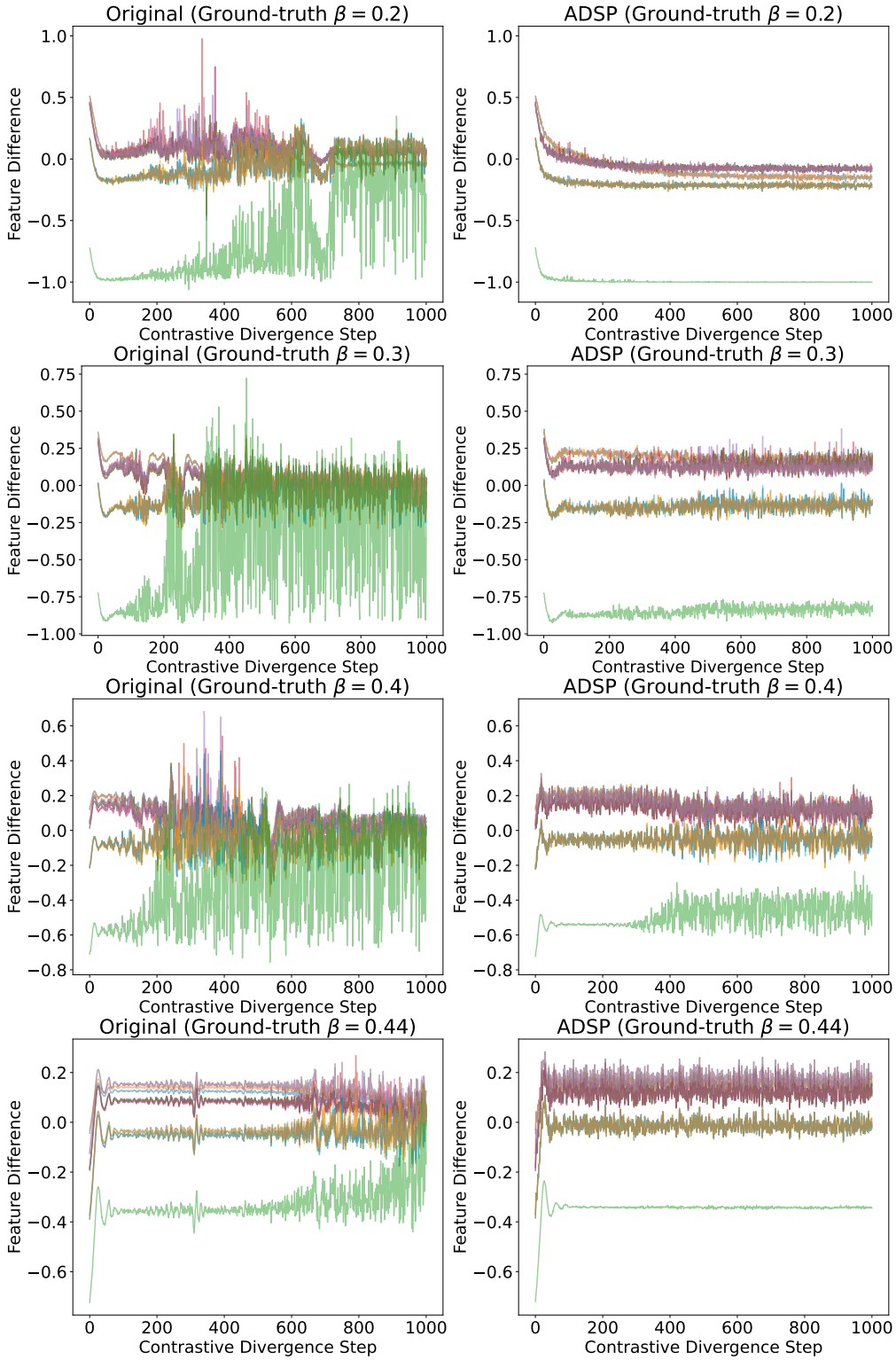


Figure A16: Feature differences $\{\langle \phi^a(G_\theta(\tilde{x}))\rangle_{\tilde{H}} - \langle \phi^a(x)\rangle_H\}_{a=1}^{13}$ of the original method (left) and ADSP (right) during the learning process with perturbed features under the Gaussian noise.

A.6 Additional Experiments on 2D Ising Model

A.6.1 Experiments on Adversarial Training, PGD Attack, and Weight Decay

We further compare the proposed ADSP with adversarial training [60] under the PGD Attack. Adversarial training is a widely-used defense scheme against adversarial attacks, while the PGD Attack is a widely-used gradient-based perturbation constructing scheme. 100% adversarial samples are used in adversarial training, while 10 attack steps and $\epsilon = 0.15$ ℓ_∞-attack budget are used in the PGD Attack. Results in Figure A17 indicate that ADSP converges smoothly to estimations $\tilde{\beta}$ that are very close to the ground-truth β, thus it is effective even under the PGD Attack, which is stronger than the Gaussian perturbation attack. However, the adversarial training cannot estimate β with a satisfactory accuracy. Besides, Figures A18, A19, A20, and A21 also show that the training process of ADSP is as stable as the adversarial training. Hence ADSP is preferable to the adversarial training in this task on statistical physics.

Weight decay is the ℓ_2 regularization on the learnable parameters, which might implicitly improve robustness to adversarial perturbations. We also conduct experiments for weight decay under the PGD Attack in Table A2. Results show that it performs poorly in this task, thus it cannot characterize and suppress the functional sensitivity of the loss function to adversarial perturbations. Therefore, a specialized defense method like ADSP with FLV is necessary to handle such adversarial noises.

Table A2: Estimated $\tilde{\beta}$ via weight decay under the PGD Attack.

Ground-truth β	0.2	0.3	0.4	0.44
Estimated $\tilde{\beta}$ (weight decay = 1e-5)	0.5911	0.2311	0.2924	0.5704
Estimated $\tilde{\beta}$ (weight decay = 5e-5)	0.7363	0.4416	0.7532	0.0716
Estimated $\tilde{\beta}$ (weight decay = 1e-4)	0.6994	0.7061	0.3775	0.0311
Estimated $\tilde{\beta}$ (weight decay = 5e-4)	0.3103	0.8537	0.1319	0.2292
Estimated $\tilde{\beta}$ (weight decay = 1e-3)	0.9418	0.4680	0.3274	0.5946

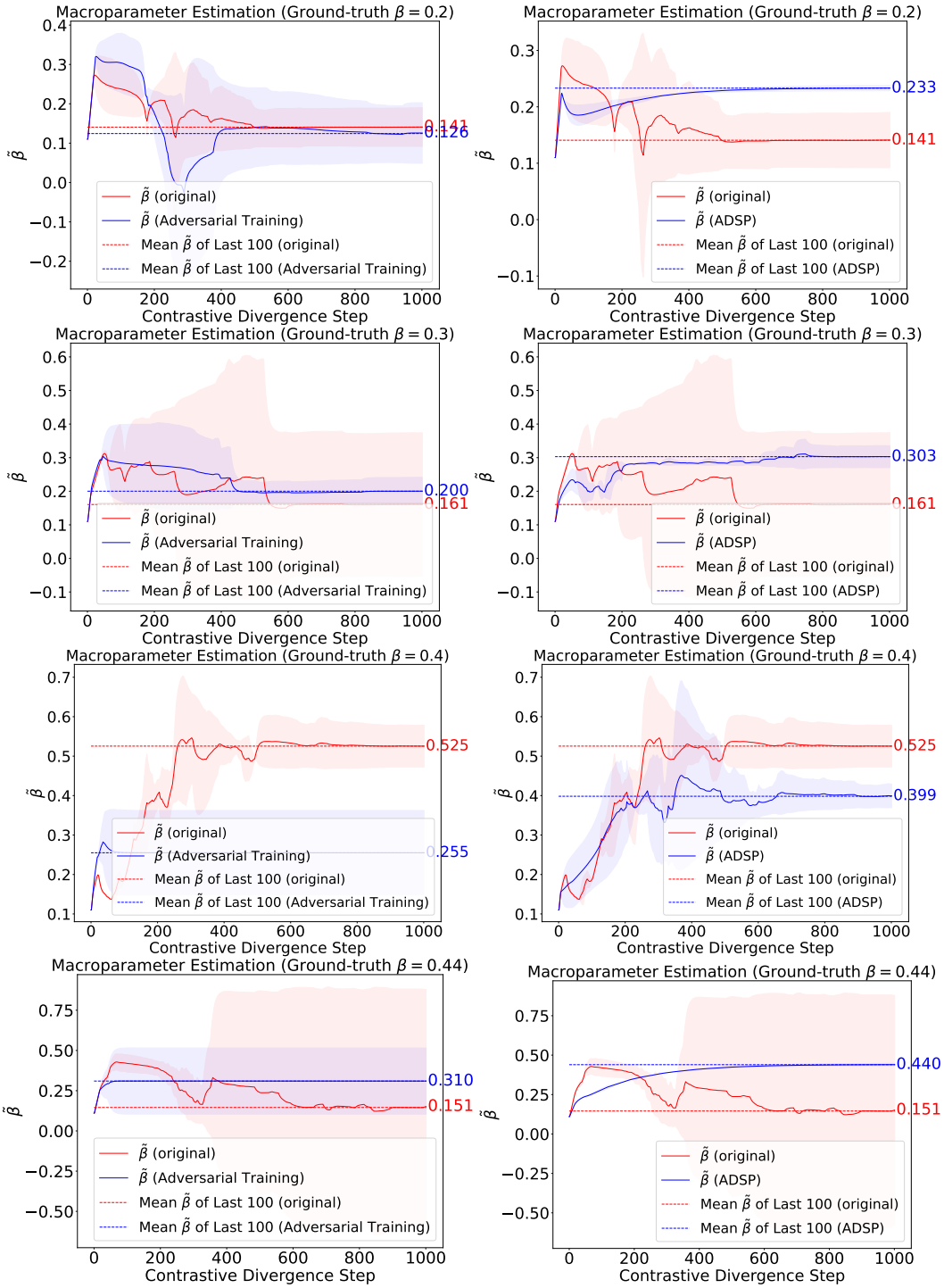


Figure A17: Learning processes of $\tilde{\beta}$ to the ground-truth β via the original method, Adversarial Training (left), and ADSP (right) with perturbed inputs under the PGD attack.

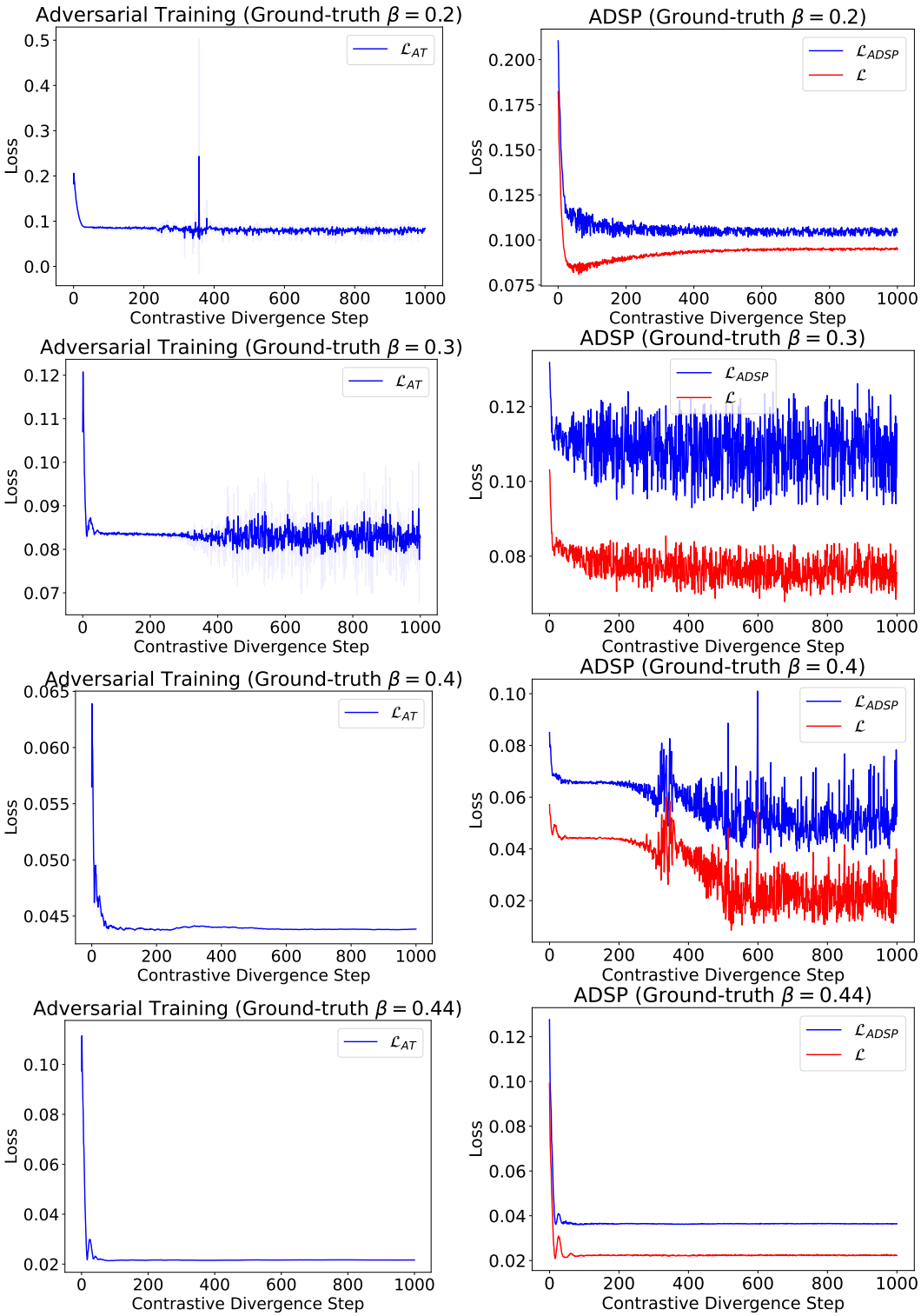


Figure A18: Loss function values of Adversarial Training (left) and ADSP (right) during the learning process with perturbed inputs under the PGD attack.

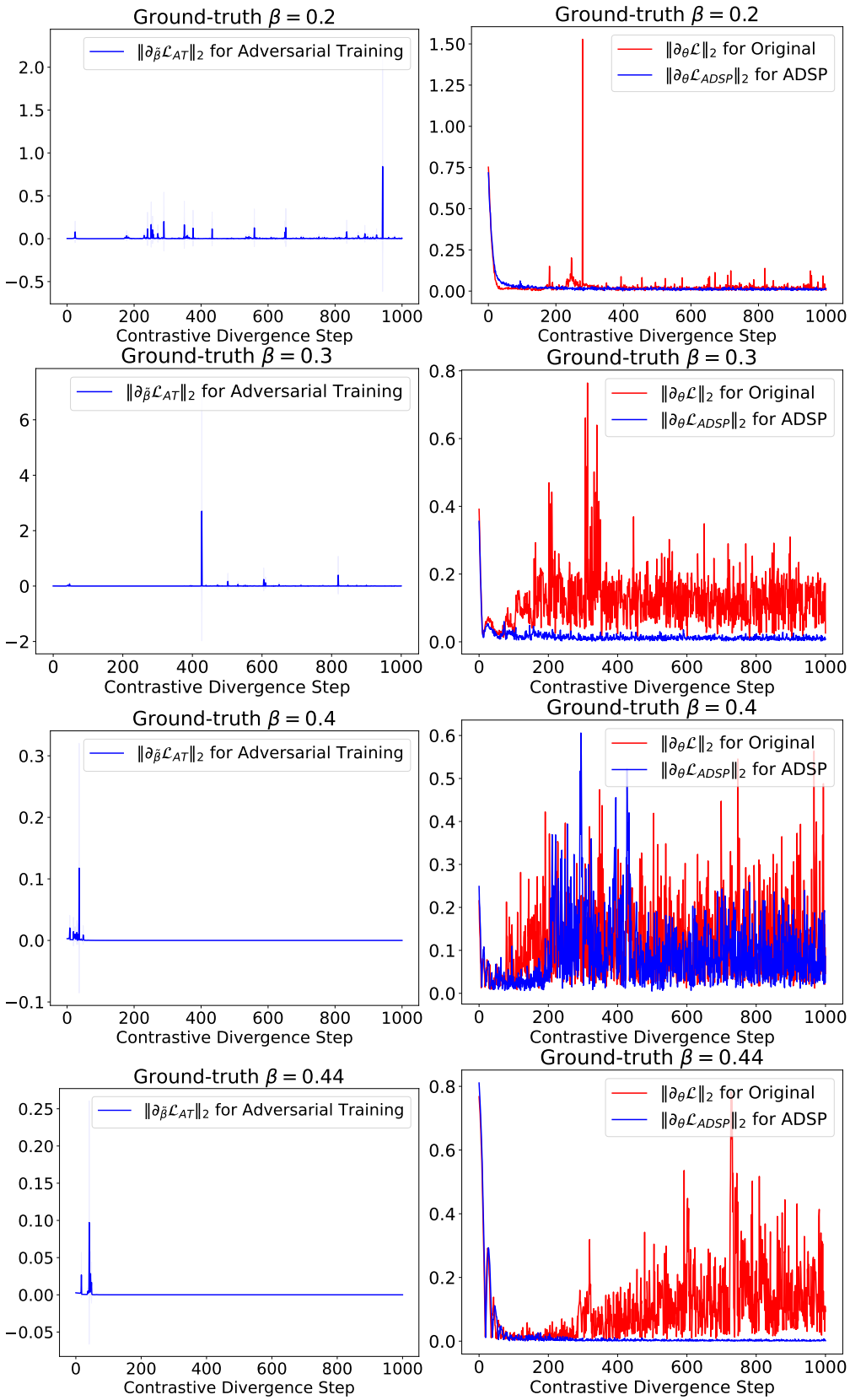


Figure A19: Gradient magnitudes of Adversarial Training (left) and ADSP (right) during the learning process with perturbed inputs under the PGD attack (left for $\partial_{\tilde{\beta}}$ and right for ∂_{θ}).

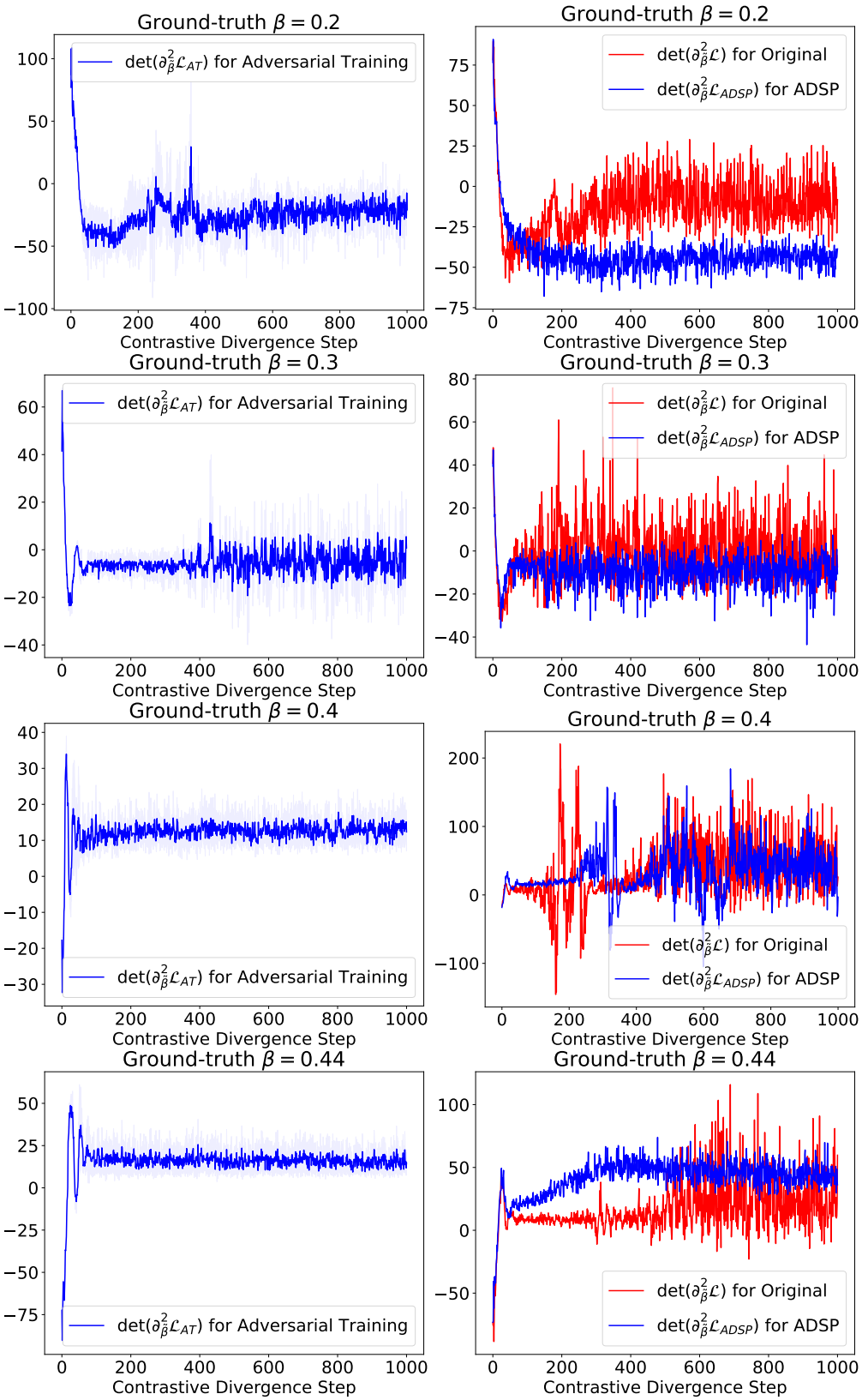


Figure A20: Hessian determinants of Adversarial Training (left) and ADSP (right) during the learning process with perturbed inputs under the PGD attack.

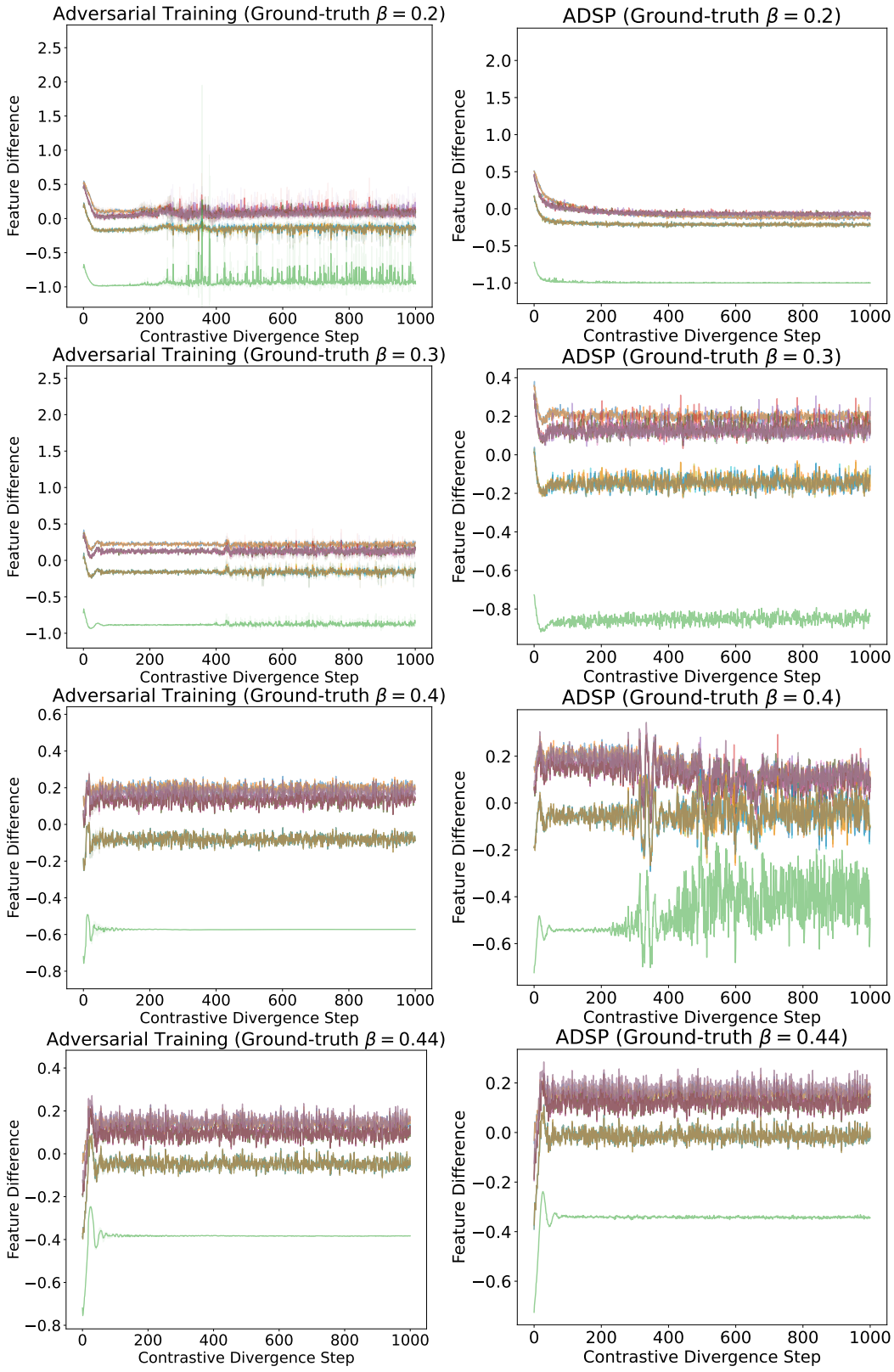


Figure A21: Feature differences $\{\langle \phi^a(G_\theta(\tilde{x}))\rangle_{\tilde{H}} - \langle \phi^a(x)\rangle_H\}_{a=1}^{13}$ of Adversarial Training (left) and ADSP (right) during the learning process with perturbed inputs under the PGD attack.

A.6.2 Defense against Stronger White-box Attacks

We also add experiments for the stronger white-box attacks C&W [9] and APGD [15], as well as the adaptive white-box attack [54] that exploits the full knowledge of ADSP, shown in Table A3. ADSP still **recovers** the physics parameter β with **high accuracies**, compared with AT.

Table A3: Defense evaluation against stronger white-box attacks across different β values and CD steps.

β	CD Step					
	0	200	400	600	800	1000
APGD+ADSP						
0.2000	0.1100	0.1220	0.1340	0.1519	0.1637	**0.1812**
0.3000	0.1100	0.2782	0.2806	0.2814	0.2817	**0.2818**
0.4000	0.1100	0.1639	0.2202	0.2748	0.3260	**0.3802**
0.4400	0.1100	0.2335	0.3482	0.3872	0.4145	**0.4374**
APGD+AT						
0.2000	0.1100	-0.1441	0.1302	0.2341	0.2531	0.2551
0.3000	0.1100	0.2906	0.0817	0.0696	0.0670	0.0663
0.4000	0.1100	0.3536	0.3537	0.3537	0.3537	0.3537
0.4400	0.1100	0.2469	0.2469	0.2469	0.2469	0.2469
C&W+ADSP						
0.2000	0.1100	0.1220	0.1400	0.1580	0.1760	**0.1939**
0.3000	0.1100	0.1989	0.2448	0.2668	0.2808	**0.3076**
0.4000	0.1100	0.2154	0.2438	0.2747	0.3049	**0.3610**
0.4400	0.1100	0.2960	0.3340	0.3582	0.3794	**0.3960**
C&W+AT						
0.2000	0.1100	0.4725	1.0768	1.0841	1.0819	1.0823
0.3000	0.1100	0.0576	0.0461	-0.0135	-0.0041	0.0098
0.4000	0.1100	0.1743	0.1738	0.1736	0.1736	0.1736
0.4400	0.1100	0.3546	0.3502	0.3484	0.3478	0.3476
Adaptive White-box Attack+ADSP						
0.2000	0.1100	0.1220	0.1401	0.1583	0.1768	**0.1957**
0.3000	0.1100	0.1784	0.2148	0.2412	0.2527	**0.2705**
0.4000	0.1100	0.1919	0.2297	0.2674	0.3172	**0.3809**
0.4400	0.1100	0.2362	0.2774	0.3244	0.3679	**0.3962**
Adaptive White-box Attack+AT						
0.2000	0.1100	-0.2006	-0.2538	-0.2091	-0.1948	-0.1784
0.3000	0.1100	0.1855	0.0632	0.0544	0.0522	0.0517
0.4000	0.1100	0.3558	0.3558	0.3558	0.3558	0.3558
0.4400	0.1100	0.3125	0.3125	0.3125	0.3125	0.3125

A.6.3 Hyperparameter Sensitivity Analysis of λ

We conduct an ablation study for the hyperparameter $\lambda = 0.08 \sim 0.12$ as follows. Results show that ADSP achieves stable performance when λ changes in a small range around 0.1.

A.6.4 Physical Mechanics of ADSP

Adding an adversarial term (regularizer) impairs the optimization on the clean loss. In standard machine learning, unregularized models achieve a lower mean clean training loss by aggressively fitting to individual training samples. However, in statistical physics pipelines, this hyper-optimization to specific MCMC sample realizations causes severe over-sensitivity to finite-sample stochastic noise.

Table A4: Performance of ADSP with different values of λ.

β \ λ	0.08	0.09	0.1	0.11	0.12
0.2000	0.1905	0.1951	0.2092	0.2161	0.1965
0.3000	0.3108	0.3157	0.2788	0.3207	0.2861
0.4000	0.3838	0.3819	0.4026	0.3723	0.3911
0.4400	0.4083	0.4299	0.4413	0.4743	0.4416

Physical Mechanics: smoother ϕ prediction leading to better β recovery.

- **MCMC Noise vs. True Feature Trajectory:** The base loss is defined as $\mathcal{L} = \sum_a (\langle \phi^a(x) \rangle_H - \langle \phi^a(G_\theta(\tilde{x})) \rangle_{\tilde{H}})^2$. In finite-lattice MCMC sampling (8×8), sample feature expectations $\langle \phi^a \rangle_{\tilde{H}}$ contain high-frequency statistical variance. Without the proposed FLV, the unregularized method over-optimizes $\mathcal{L}$ by chasing these random sample-level statistical fluctuations, which yields a slightly lower average loss value $\mathcal{L}$. However, this forces the learned macroparameter $\tilde{\beta}$ into narrow and spurious local minima that do not reflect the true thermodynamic state β^*.
- **FLV as a Feature-Space Smoothness Regularizer:** The FLV term $F(\tilde{\beta}, \theta) = \|\partial_{\phi^a} \mathcal{L}\|_2$ quantifies the local loss sensitivity w.r.t. the key physical feature manifold ϕ^a. In thermodynamics, physical observables vary smoothly as a function of feature expectations away from phase transitions. By penalizing $\|\partial_{\phi^a} \mathcal{L}\|_2$, ADSP flattens the local loss landscape around the physical feature manifold ϕ^a.
- **Gradient Backpropagation and Unbiased Trajectory:** As derived in (18) and (19) , the subdifferential gradient update for $\tilde{\beta}$ relies on backpropagation through feature sensitivities:

$$\partial_{(\tilde{\beta},\theta)} F(\tilde{\beta}, \theta) = \partial_{(\tilde{\beta},\theta)} \phi \cdot \partial_\phi \|\partial_\phi \mathcal{L}(\phi)\|_2. \tag{28}$$

  By constraining the derivative magnitude $\partial_\phi \mathcal{L}$, ADSP suppresses the high-frequency derivative spikes caused by the MCMC noise. As a result, the stochastic gradient updates for $\tilde{\beta}$ become smooth and consistent. Instead of chasing local batch noise to lower $\mathcal{L}$, $\tilde{\beta}$ follows a smooth macrostate trajectory toward the true thermodynamic fixed point β^*.

This physical mechanism is directly supported by our diagnostic metrics in the experiments:

- **Gradient Magnitude (Figures A4 and A9):** $\|\partial_{\tilde{\beta}} \mathcal{L}_{ADSP}\|_2$ and $\|\partial_\theta \mathcal{L}_{ADSP}\|_2$ show fewer spikes and decay smoothly compared to the unregularized method.
- **Hessian Determinant (Figures A5 and Figures A10):** The Hessian determinant $\det(\partial^2_{\tilde{\beta}} \mathcal{L}_{ADSP})$ is more steady compared to the unregularized method, which indicates that ADSP eliminates ill-conditioned local minima and constructs a well-behaved basin of attraction around β^*.
- **Feature Difference Consistency (Figures A6 and A11):** While the feature differences of the original method fluctuates significantly, ADSP maintains steady feature matching across steps, which leads to dataset-level thermodynamic consistency.

Societal Impact

This research strictly adheres to the ethical guidelines for security and privacy research, prioritizing responsible methodology and disclosure. The primary goal of this work is fundamentally defensive: to uncover and understand the vulnerabilities within current adversarial machine learning systems and to inform the design of more robust countermeasures.

All experiments are conducted exclusively in controlled, academic environments. The data sets used in this study are publicly available, widely-used in the academic community, and conform to their respective licenses. No personally identifiable information or sensitive private data is collected, stored, or processed throughout the course of this research.

As this research focuses solely on technical system analysis and publicly available data, it does not involve human subjects.

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